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AMPANet: Adaptive Multi-scale Pairwise Attention for Remote Sensing Image Scene Classification

Zhenghua Huang, *Member, IEEE*, Yingjie Zhang, Qian Li, Yaozong Zhang, Chunhua Deng, Xin Xu, *Senior Member, IEEE*, and Yansheng Li, *Senior Member, IEEE*

***Abstract*—Remote sensing scene classification (RSSC) has been one of the most important research topics in Earth observation applications. Although CNN-based methods have been extensively employed to learn hierarchical visual features, their limited receptive fields make long-range dependencies and complex object relationships difficult to capture. Although Transformer-based methods have improved global interaction modeling, they still insufficiently exploit multi-scale semantic relationships. To address these limitations, this paper proposes an Adaptive Multi-scale Pairwise Attention Network (AMPANet), in which the core component of AMPANet is an Adaptive Multi-scale Pairwise Attention Module (AMPAM) that jointly models multi-scale contextual information and pairwise semantic interactions. Specifically, Multi-Scale Depthwise Separable Convolution (MDWC) is designed to efficiently extract multi-scale features, while Dynamic Coordinate Pooling (DCP) is incorporated to enhance directional spatial perception. In addition, Multi-Head Adaptive Pairwise Attention (MAPA) is developed to explicitly model pairwise interactions among semantic regions, and Adaptive Feature Modulation (AFM) is employed to dynamically integrate multi-scale relational features while suppressing redundant information. By jointly considering multi-scale representation learning, directional context modeling, and object-level relational interaction, both discriminability and robustness of remote sensing scene representations are effectively improved. Quantitative associated with qualitative results of experiments on four benchmark datasets (UCM, AID, NWPU RESISC45 and MEET50) demonstrate that our AMPANet is effective and achieves more superior classification performance than state of-the-art (SOTA) methods, particularly in complex multi-scale remote sensing scenes.**

***Index Terms*—Remote sensing scene classification (RSSC), adaptive multi-scale pairwise attention module (AMPAM), dynamic coordinate pooling (DCP), adaptive feature modulation (AFM), multi-scale feature learning.**

I. Introduction

REMOTE sensing scene classification aims to automatically assign semantic labels to aerial or satellite images according to their land-cover and land-use characteristics [1, 2]. As one of the fundamental tasks in remote sensing image understanding, scene classification serves as a crucial component in numerous practical applications, including urban planning, environmental monitoring, agricultural assessment, disaster management, and military reconnaissance [3]. With the rapid advancement of Earth observation technologies, modern remote sensing platforms can acquire images with increasingly high spatial, spectral, and temporal resolutions [4]. Consequently, vast quantities of remote sensing data are continuously generated, creating an urgent demand for intelligent and automated scene interpretation methods [5, 6].

Traditional scene classification methods mainly rely on handcrafted feature descriptors, such as Scale-Invariant Feature Transform (SIFT) [7], Histogram of Oriented Gradients (HOG) [8], Generalized Search Tree (GIST) [9], and Texture Descriptor (TD) [10]. Although these methods can capture low-level texture and structural information, their representation capability is limited when dealing with complex scene semantics and large intra-class variations. Consequently, they often exhibit poor generalization performance on large-scale and diverse remote sensing datasets. The emergence of deep learning [11, 12] has significantly advanced remote sensing scene classification. Convolutional Neural Networks (CNNs) [13], including AlexNet [14], ResNet [15], and DenseNet [16], have demonstrated remarkable performance by learning hierarchical semantic representations directly from raw images. However, CNN-based approaches still suffer from several inherent limitations. Owing to the local receptive field of convolution operations, they are less effective in capturing long-range contextual dependencies. Moreover, remote sensing scenes usually contain objects with substantially different scales, making fixed-scale feature extraction insufficient for representing diverse object distributions. In addition, scene semantics are often determined by the interactions among multiple objects rather than individual objects alone, while conventional CNNs mainly focus on local appearance information and neglect such relational cues.

Recently, vision Transformers (ViTs) have attracted considerable attention, unlike CNNs, Transformers can directly establish long-range interactions between arbitrary spatial locations, enabling more comprehensive scene understanding. Transformer-based methods with different attentions (*e.g.,* coordinate attention (CA) [17] and cross attention [18]) have achieved promising performance in various remote sensing

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Zhenghua Huang, Chunhua Deng, and Xin Xu are with School of Computer Science and Technology, Wuhan University of Science and Technology, Wuhan 430065, China, and Zhenghua Huang is also with the Technology Innovation Center for Collaborative Applications of Natural Resources Data in GBA, MNR, Guangzhou 510075, China. (e-mails: zhhuang@wust.edu.cn, dchzx@wust.edu.cn, and xuxin@wust.edu.cn)

Yingjie Zhang and Yaozong Zhang are with School of Electrical and Information Engineering, Wuhan Institute of Technology, Wuhan 430205, China. (e-mails: 22403010030@wit.edu.cn and zhangyaozong@wit.edu.cn)

Qian Li is with the School of Electronic Information Engineering, Wuhan Donghu University, Wuhan 430212, China (e-mail: liqian@wdu.edu.cn).

Yansheng Li is with School of Remote Sensing and Information Engineering, Wuhan University, Wuhan 430079, China, and also with the Technology Innovation Center for Collaborative Applications of Natural Resources Data in GBA, MNR, Guangzhou 510075, China. (e-mails: yansheng.li@whu.edu.cn)

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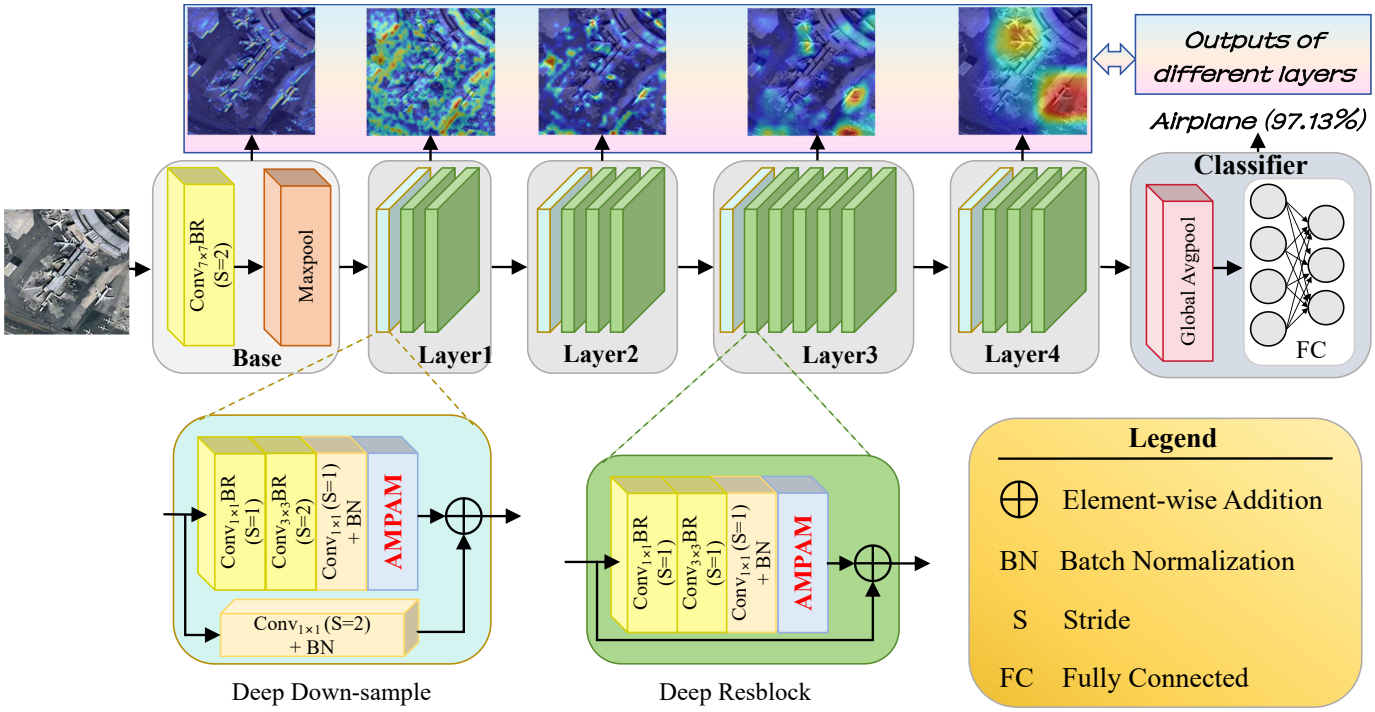
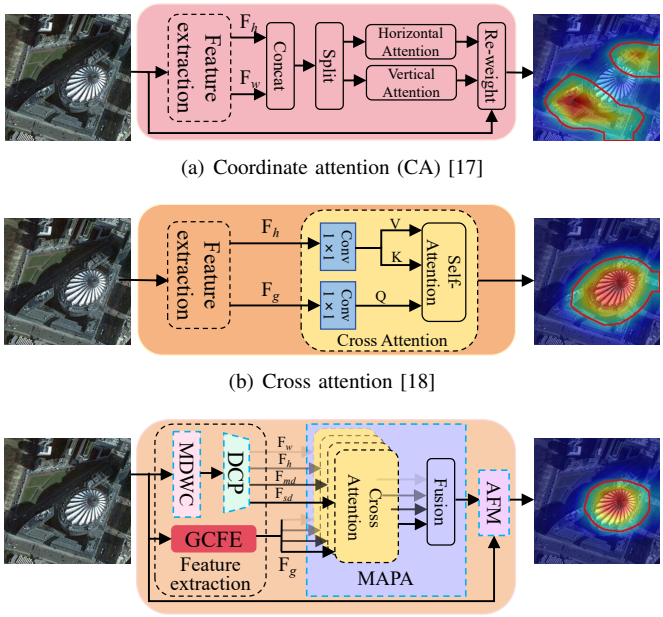


Fig. 1. The overall framework of our adaptive multi-scale pairwise attention network (AMPANet).



(a) Coordinate attention (CA) [17]

(b) Cross attention [18]

(c) Our AMPAM

Fig. 2. Visual results with different attentions.

tasks, including scene classification, object detection [19], and semantic segmentation [20, 21]. Nevertheless, standard self-attention mechanisms generally incur high computational complexity when processing high-resolution imagery and often fail to emphasize the most discriminative semantic relationships among scene components [22]. Furthermore, most attention mechanisms operate on a single feature scale, making them less effective in handling significant variations in object size and scene structure.

To overcome these limitations, multi-scale representation learning has emerged as an important research direction in remote sensing scene classification. Multi-scale approaches aim to capture complementary information from different spatial resolutions, enabling networks to simultaneously perceive fine-grained local details and coarse-grained global context [23, 24]. Representative strategies include multi-branch architectures [25–27], feature pyramid networks [28, 29], dilated convolutions [30], and hierarchical Transformer structures [31]. Although these methods enhance feature diversity, most existing approaches employ simple feature aggregation techniques such as concatenation or summation. Such fusion strategies cannot dynamically adapt to varying scene characteristics and may introduce redundant or irrelevant information into the final representation. Another promising research direction involves attention-based relational modeling [32]. Existing attention mechanisms generally focus on channel attentions (such as convolutional block attention module (CBAM) [33] and efficient channel attention (ECA) [34]), spatial attention (*e.g.*, squeeze-and-excitation (SE) [35] and efficient multi-scale attention (EMA) [36]), or global self-attention [37]. While these techniques improve feature discrimination by emphasizing informative regions or channels, they often overlook explicit pairwise interactions between semantic entities. In complex remote sensing scenes, objects rarely exist independently. Instead, their spatial configurations and contextual relationships collectively determine scene semantics.

To address these issues, a novel network, whose architecture is shown in Fig. 1, is reported for remote sensing scene classification, in which the key component is an Adaptive Multi-scale Pairwise Attention Module (AMPAM). Figure 2 provides a visual comparison of our network with different attentions, from which it can be seen that our AMPAM is more beneficial for highlighting the targets, thereby improving the

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overall scene representation ability. The main contributions of this work are summarized as follows:

- A novel Adaptive Multi-scale Pairwise Attention Module (AMPAM) that jointly exploits multi-scale contextual information and pairwise semantic relationships for remote sensing scene classification.
- A pairwise attention mechanism is developed to explicitly model interactions among semantic regions, enabling the extraction of more discriminative scene representations.
- An adaptive multi-scale fusion strategy is designed to dynamically aggregate complementary features from different spatial scales, improving robustness to object-scale variations and scene complexity.

II. Related works

A. Multi-Scale Feature Learning for RSSC

Multi-scale representation learning has become an essential research direction in remote sensing scene understanding. Remote sensing images inherently contain objects with significant scale variations, so effective scene representation requires the ability to simultaneously capture local details and global contextual information. To address this challenge, numerous multi-scale learning strategies have been proposed.Bai et al. [38] employed Octave convolution and a feature fusion module to extract and integrate multi-frequency and multi-scale information to improve the ability to distinguish complex scenes (MF2CNet). Zhao et al. [39] utilized gradient to steer the fusion of multi-scale features, addressing issues related to scale variation and the precise localization of key objects in remote sensing scene classification (GMFANet). Yang et al. [40] designed a Semantically Aware Graph Network (SAGN), which adaptively identifies multiple semantic regions in an image and refines pixel features via a novel region-based graph convolution, effectively capturing both local semantic information and global contextual information to improve classification accuracy. Wang et al. [41] proposed HRNetv2, a high-resolution network that maintains high-resolution representations throughout the network and performs parallel multi-scale fusion across sub-networks, achieving powerful multi-scale feature representation. Hu et al. [42] proposed a Diffusion Self-Distillation Network (DSDNet), which incorporates conditional diffusion denoising into the self-distillation framework to suppress disruptive information in shallow features and facilitate more reliable knowledge transfer from deep semantic features, thereby improving robust feature representation for remote sensing scene classification. Sitaula et al. [43] developed an enhanced multi-level feature learning framework by combining an Enhanced Attention Module (EAM), atrous spatial pyramid pooling (ASPP), and global average pooling to refine features at different network levels, followed by multi-level feature fusion to improve the robustness and discriminability of remote sensing scene representations. Although these methods enhance scale perception, they often lack deep synergy across different scale levels and cannot explicitly decouple complex multi-directional contexts, thereby limiting their adaptability in complex remote sensing scenarios.

Moreover, in the context of multi-level feature fusion, bridging the inherent semantic gaps and distribution discrepancies across different feature levels is equally crucial. Zhao et al. [44] introduced EMSCNet, leveraging multi-sample contrastive learning and knowledge distillation to transfer discriminative knowledge to the backbone, thereby improving inter-class separability and intra-class consistency. Such methods relied on data augmentation or loss function constraints, and could not adaptively address the distribution differences among multiple features during feature fusion.

B. Transformer-Based Scene Representation Learning

Recently, Transformer architectures have emerged as a powerful alternative to CNNs in visual recognition tasks. Inspired by their remarkable success in natural language processing, Dosovitskiy et al. [37] successfully applied a pure Transformer architecture to image classification for the first time, demonstrating performance comparable to CNNs on large-scale datasets by processing images as sequences of patches. Chen et al. [18] proposed a dual-branch Vision Transformer that processes image patches of different sizes separately and fuses features from both branches via an efficient cross-attention module, effectively learning multi-scale visual representations and improving classification accuracy. Liu et al. [45] introduced a hierarchical structure and a shifted window mechanism, whose hierarchical design builds multi-scale feature maps and the shifted windows facilitate cross-window connections while confining computation to non-overlapping local windows, significantly improving model efficiency and performance. Yuan et al. [46] proposed Tokens-to-Token ViT, which models local structural information through a progressive "tokenization" process and incorporates a deep and narrow backbone design, enhancing both the robustness and computational efficiency of ViT. Li et al. [47] proposed a Swin Transformer-based framework for joint underwater image enhancement and captioning. The framework integrates grid-level features with spatial cluster representations through multi-head cross-attention and jointly optimizes visual restoration and semantic description generation via multisupervised chained task learning. Zhai et al. [48] introduced a vision Transformer paradigm (A-ViT) by incorporating a local–global attention mechanism, it adaptively selects a subset of image patches for fine-grained global interaction while aggregating the rest locally, achieving an excellent balance between accuracy and speed. Liu et al. [49] combined CNN and Transformer models to effectively extract global structural features from target scene images, all while maintaining a relatively low network parameter overhead. Most of these methods adopt a single information flow and a globally-centered modeling paradigm, which can easily dilute the crucial local directional spatial structures in remote sensing scenes, leading to confusion when dealing with scenes where local targets are similar but their spatial layouts differ.

C. Attention Mechanisms and Relational Modeling

Attention mechanisms have become a core component of modern deep learning architectures due to their ability to selectively emphasize informative features. In remote sensing scene

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classification, attention mechanisms are widely employed to improve feature discrimination and semantic representation by modeling dependencies among channels, spatial regions, and multi-level features. Chen et al. [50] designed a Multi-branch Local Attention Network (MBLANet), integrating a convolutional local attention module into the ResNet backbone. This module uses convolutional channel attention and local spatial attention in parallel to refine channel weights and perceive semantic similarities within spatial neighborhoods, respectively, thereby improving local relational representation and feature distinguishability. Li et al. [51] proposed AquaSketch-enhanced underwater image captioning, which strengthens sketch cues while preserving detail and background information. A top-down dual-branch pyramid integrates multi-scale features through attention-based cross-scale fusion, providing complementary object representations for caption generation. Wan et al. [52] took a different approach by proposing a Hyperparameter-Free Attention Module (HFAM), which avoids complex convolution or fully connected operations and adaptively generates hybrid attention weights through simple statistical computations on feature maps, achieving lightweight and efficient plug-and-play functionality. Ma et al. [53] constructed a Multi-scale Sparse Cross-Attention Network (MSCN), whose attention mechanism design more explicitly incorporates relational modeling. It first employs multi-scale sparse cross-attention to capture cross-level feature interactions through multi-scale pooling and sparse operations, thereby filtering redundancy and fusing features across levels. Then, the grouped CBAM enhancer is used to enhance the fused features from local to global, further highlighting key information for classification. Most of these methods focus on saliency modeling of local regions or implicit dependency learning, but fail to fully explore the correlation between global semantics and local details.

III. Methodology

In this section, we introduce our AMPANet in detail. To be specific, we first introduce the overall architecture of the network. Then, we provide the detail introduction of our AMPAM which is the core component used in both deep down-sample and residual blocks.

A. Overall Architecture

The overall architecture of our network is shown in Fig. 1, in which the backbone is ResNet50 [15]. Specifically, our AMPANet is composed of the base layer including a "convolution (Conv, whose kernel size is 7×7), batch normalization (BN), and rectified linear unit (ReLU)" ($\text{Conv}_{7\times7}\text{BR}$) block and a maximum pooling (Maxpool) for low-level visual information extraction [54], intermediate layers which are the core procedures and is composed of deep down-sample block and deep resblock, and a classifier involving in a global average pooling (Avgpool) and a fully connected layer. In both deep down-sample block and deep resblock, our AMPAM is embedded to ensure that multi-scale feature fusion and interaction are seamlessly integrated at each feature extraction stage, enabling the network to establish a synergistic enhancement mechanism between local directional features and global semantics. In the following, we mainly describe the architectures of two blocks.

1) Deep Down-sample: The module is responsible for reducing the spatial resolution of feature maps and expanding the receptive field during the transition between various stages of the network, thereby extracting higher-level global semantic information. Specifically, its main branch first performs dimensionality reduction by $\text{Conv}_{1\times1}\text{BR}$ with a stride of 1, then halves the length and width using $\text{Conv}_{3\times3}\text{BR}$ with a stride of 2, doubles the number of channels through $\text{Conv}_{1\times1}\text{BN}$ (stride = 1), and embeds our AMPAM at the end of the main branch. Meanwhile, $\text{Conv}_{1\times1}\text{BN}$ (stride = 2) is employed by a residual branch to ensure strict dimensionality matching with the output of the main branch.

2) Deep Resblock: To further deepen the network layers while maintaining the spatial dimensions and channel count of the feature map unchanged, we design a deep resblock to extract deep-level nonlinear features and refine their complex spatial relationships. The main branch of the deep resblock consists of $\text{Conv}_{1\times1}\text{BR}$ (stride = 1), $\text{Conv}_{3\times3}\text{BR}$ (stride = 1), $\text{Conv}_{1\times1}\text{BN}$ (stride = 1), and AMPAM, whose output is element-wisely added by the input for the result of the deep resblock.

It is worth noting that AMPAM is embedded in both blocks and serves as the key component of the network, as detailed below.

B. Adaptive Multi-scale Pairwise Attention Module (AMPAM)

As the core innovation of our network, AMPAM is designed as a "plug-and-play" module for multi-scale feature fusion and interaction. As shown in Fig. 3, our AMPAM processes the input features mainly through four sequential components: firstly, Multi-Scale Depthwise Separable Convolution (MDWC) based on dynamic kernel selection to efficiently extract and fuse local spatial information from different receptive fields, Dynamic Coordinate Pooling (DCP) to capture structural features in the horizontal and vertical, main diagonal and secondary diagonal dimensions, Multi-Head Adaptive Pairwise Attention (MAPA) to achieve deep feature interaction between these two orthogonal directions and the global context, and Adaptive Feature Modulation (AFM) to generate dynamic scaling and shifting parameters, complete precise recalibration and deep fusion, thereby enhancing feature representation capability while ensuring stable gradient propagation. In the following, we describe their internal structures in detail.

1) Multi-Scale Depthwise Separable Convolution (MDWC): For conventional CNNs, the single-branch convolution architecture is impossible to capture multi-scale features, primarily due to the constraints of fixed kernel sizes and local receptive fields. To address this issue, we propose a multi-scale depthwise separable convolution (MDWC), which uses a dynamic convolution kernel aggregation (DCKA) to generate weights for adaptively evaluating and aggregating the outputs of depthwise separable convolution branches with different receptive fields.

To be specific, for the input feature map F_{in}, the MDWC contains two parallel branches: one is that the input feature

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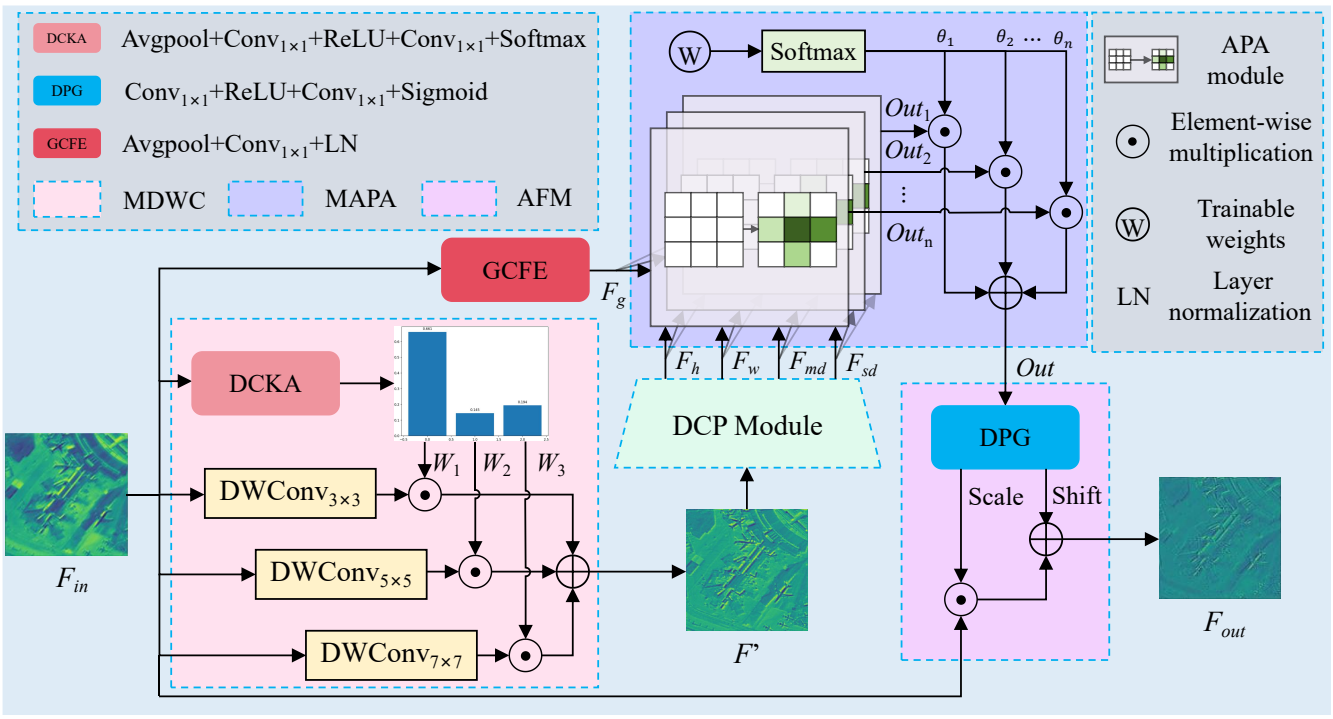

Fig. 3. The internal architecture of our AMPAM.

map F_{in} is processed by DCKA to produce three adaptive weights

$$\{W_1, W_2, W_3\} = DCKA\left(F_{in}\right), \tag{1}$$

where W_1, W_2 and W_3 represent the weights corresponding to the first, second, and third DWConv, respectively. DCKA is composed of "Avgpool + $\text{Conv}_{1\times1}$ + ReLU + $\text{Conv}_{1\times1}$ + Softmax", where Avgpool$(\cdot)$ is an average pooling operator, $\text{Conv}_{1\times1}$ is a 1×1 Conv (if the stride of subsequent convolution is not specified, its default value is 1), ReLU$(\cdot)$ is a ReLU activation function, and Softmax$(\cdot)$ is a softmax activation function.

The other is the multi-scale feature extraction module which includes three DWConv with kernel sizes of 3×3, 5×5, and 7×7 to capture local details and contextual information under different receptive fields. Then, the features extracted from different convolutions are multiplied by their corresponding adaptive weights from DCKA and summed to obtain the intermediate feature map

$$F' = MDWC\left(F_{in}\right) = \sum_{i=1}^{3} W_i \odot DWConv_{2\times i+1}\left(F_{in}\right), \tag{2}$$

where W_i represents the weights corresponding to the i-th $DWConv_{2\times i+1}(\cdot)$ and $\odot$ represents the element-wise multiplication.

2) Dynamic Coordinate Pooling (DCP): After obtaining the intermediate feature map F', a Dynamic Coordinate Pooling (DCP) module is introduced to further enhance the directional perception ability of multi-scale features. Different from traditional global pooling for compressing spatial information into a single global descriptor, DCP performs adaptive pooling along different directions to preserve more positional and directional information, so it is beneficial for remote sensing scenes containing objects with diverse orientations and spatial layouts. Its overall structure (shown in Fig. 4) is designed to transform the input feature map into four direction-aware coordinate features, including the horizontal feature F_h, vertical feature F_w, main diagonal feature F_{md}, and secondary diagonal feature F_{sd}, which can be formulated as

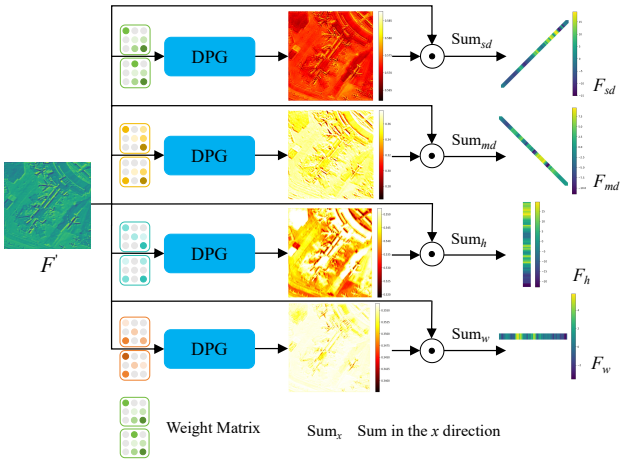

Fig. 4. Dynamic Coordinate Pooling (DCP).

$$\{F_h, F_w, F_{md}, F_{sd}\} = DCP\left(F'\right). \tag{3}$$

Specifically, given the input feature map F', DCP first employs a parallel dynamic parameter generator (DPG) (which is composed of "$\text{Conv}_{1\times1}$ + ReLU + $\text{Conv}_{1\times1}$ + Sigmoid") to

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generate direction-specific weight matrices. Since the parallel sub-networks are independent networks, their parameters are randomly initialized, resulting in the generated weight matrices being different from each other. These weight matrices correspond to the horizontal, vertical, main diagonal, and secondary diagonal directions, respectively. After obtaining the direction-specific weight matrices, each weight matrix is first multiplied with the input feature map in an element-wise manner to emphasize informative spatial positions and suppress less relevant regions. Then, the weighted feature maps are summed along their corresponding directions to generate four direction-aware coordinate features. For example, the coordinate feature in the x direction can be formulated as

$$F_x = \mathrm{Sum}_x(DPG(F') \odot F') \tag{4}$$

Here, Sum_x denotes summation along the spatial dimension associated with direction x, where $x \in \{h, w, md, sd\}$. More specifically, for a weighted feature map $Z \in \mathbb{R}^{C\times H\times W}$, i and j denote the height and width indices, respectively. $\mathrm{Sum}_h(Z)_{c,i} = \sum_{j=1}^{W} Z_{c,i,j}$ aggregates each horizontal row, whereas $\mathrm{Sum}_w(Z)_{c,j} = \sum_{i=1}^{H} Z_{c,i,j}$ aggregates each vertical column. Similarly, Sum_{md} and Sum_{sd} aggregate diagonal positions with constant $i-j$ and constant $i+j$, respectively. Therefore, this operator performs one-dimensional directional aggregation rather than summation over both spatial dimensions.

For non-square feature maps ($H \neq W$), the diagonal aggregation operators are applied directly to the original $H \times W$ grid. Using the one-based indices defined above, each position (i, j) is mapped to a main-diagonal bin $t_{md} = i - j + W$ and a secondary-diagonal bin $t_{sd} = i + j - 1$. Both indices range from 1 to $H + W - 1$, and every position contributes to exactly one bin in each direction. In implementation, the direction-specific weighted feature maps are flattened, and `scatter_add` accumulates features sharing the same bin index within each direction. Only valid positions within the grid are included, naturally accommodating diagonals of different lengths without padding, cropping, or interpolation. Consistent with Eq. 4, the default implementation sums weighted features without normalization by diagonal length, yielding F_{md} and F_{sd} of size $C \times (H + W - 1)$. For example, a 3×5 feature map produces seven bins in each direction, containing $1, 2, 3, 3, 3, 2, 1$ positions, respectively. These aggregation operators can be written as

$$\begin{aligned} \mathrm{Sum}_{md}(Z)_{c,t_{md}} &= \sum_{\substack{1\le i\le H,\ 1\le j\le W \\ i-j+W=t_{md}}} Z_{c,i,j}, \\ \mathrm{Sum}_{sd}(Z)_{c,t_{sd}} &= \sum_{\substack{1\le i\le H,\ 1\le j\le W \\ i+j-1=t_{sd}}} Z_{c,i,j}. \end{aligned} \tag{5}$$

3) Multi-Head Adaptive Pairwise Attention (MAPA): To improve the model's discriminative ability, it is necessary to search for the correlations between features in four directions and global contextual information. Unfortunately, it is difficult to fully express their complex spatial dependencies using a simple concatenation or addition, and will produce a problem of feature mismatch during fusion. To cope with this problem,

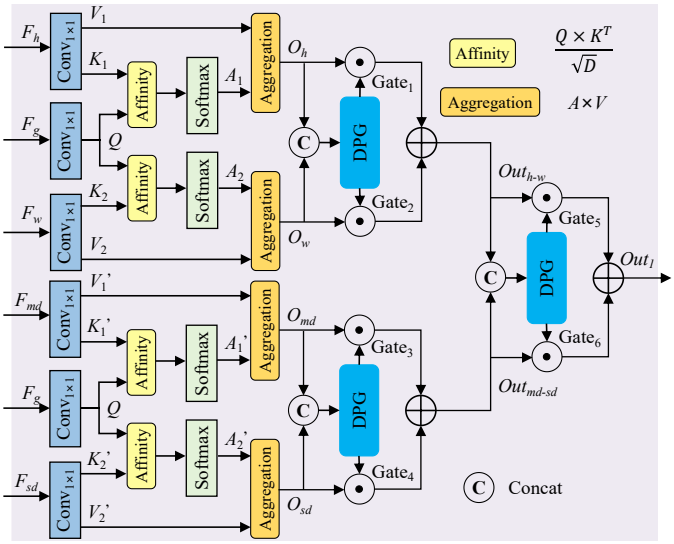

Fig. 5. One attention head in the Multi-Head Adaptive Pairwise Attention (MAPA) module.

a multi-head adaptive pairwise attention (MAPA) is reported to adaptively adjust the fusion weights of multiple features, thereby ensuring that the network suppresses irrelevant noise while explicitly highlighting the most discriminative spatial information. The output of our MAPA is

$$Out = MAPA\left(F_g, F_h, F_w, F_{sd}, F_{md}\right), \tag{6}$$

where F_h, F_w, F_{sd} and F_{md} are obtained from DCP while F_g is obtained from a global context feature extraction (GCFE) module, which is calculated as

$$F_g = GCFE\left(F_{in}\right), \tag{7}$$

where $GCFE(\cdot)$ is a module for global context feature extraction, consisting of "Avgpool + $\mathrm{Conv}_{1\times1}$ + LN", $\mathrm{LN}(\cdot)$ is layer normalization.

To more clearly explain the internal computational logic of the MAPA module, we provide its detail computation procedure of one of the attention heads. For each attention head, its architecture is shown in Fig. 5, which has two branches. Take the first attention head as an example, we can see from the figure that, in the upper branch, the global features F_g as well as the directional features F_h and F_w are mapped into new features using a 1×1 convolution, respectively named as Q, K_1 and V_1, K_2 and V_2. Let B denote the batch size, C the common channel number of the global and directional features, and n the number of heads, with $D = C/n$. The input sequences are $F_g \in \mathbb{R}^{B\times C\times 1}$ and $F_x \in \mathbb{R}^{B\times C\times L_x}$, where L_x includes L_h, L_w, L_{sd} and L_{md}, $L_h = H$, $L_w = W$, $L_{md} = L_{sd} = H + W - 1$. The query projection preserves size $B \times C \times 1$, while the key and value projections preserve size $B \times C \times L_x$. Head splitting gives $B \times n \times D \times 1$ for Query and $B \times n \times D \times L_x$ for both Key and Value. Then, we can calculate the attention map as

$$A_i = \mathrm{softmax}\left(\frac{Q \times K_i^T}{\sqrt{D}}\right), \quad i \in \{1, 2\}, \tag{8}$$

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where D is the dimension of Q. For the matrix multiplications in Eqs. 8, the per-head representations are $Q \in \mathbb{R}^{B\times 1\times D}$, $K^T \in \mathbb{R}^{B\times D\times L_x}$, and $V \in \mathbb{R}^{B\times L_x\times D}$, using the appropriate transposes of the stored tensors. After that, we aggregate the V using the attention weights to obtain attention outputs,

$$O_h = A_1 \times V_1, O_w = A_2 \times V_2. \tag{9}$$

For each head, $Q \times K^T$ and its softmax output have size $B \times 1 \times L_x$. Multiplication by V yields $B \times 1 \times D$, which is transposed to $B \times D \times 1$.

After obtaining the horizontal and vertical features guided by the global features O_h and O_w, we further introduce a gating mechanism to adaptively fuse the directional information from these two branches. Specifically, the horizontal and vertical features are first concatenated to provide joint directional context. Then, the DPG module is used to generate two adaptive gate weights for the two directional features:

$$\{\text{Gate}_1, \text{Gate}_2\} = DPG\left(Cat\left(O_h, O_w\right)\right), \tag{10}$$

where Gate_1 and Gate_2 are the adaptive weights corresponding to O_h and O_w, respectively, and $Cat(\cdot)$ represents the concatenation operation. With the help of the generated gate weights, the horizontal and vertical features are then adaptively fused by weighted summation. The final output Out_{h-w} of the horizontal–vertical attention branch can be formulated as

$$Out_{h-w} = \text{Gate}_1 \odot O_h + \text{Gate}_2 \odot O_w, \tag{11}$$

which contains both horizontal and vertical directional information.

Similarly, in the down branch of Fig. 5, we map F_{md} and F_{sd} to generate keys and values ($K_1^{'}$ and V_1' for F_{md}, and K_2' and V_2' for F_{sd}). Then, we can obtain O_{md} and O_{sd} with the employment of Eqs. 8 and 9. Finally, we obtain the output map Out_{md-sd} of main–secondary diagonal attention branch,

$$\begin{aligned} \{\text{Gate}_3, \text{Gate}_4\} &= DPG\left(Cat\left(O_{md}, O_{sd}\right)\right) \\ Out_{md-sd} &= \text{Gate}_3 \odot O_{md} + \text{Gate}_4 \odot O_{sd} \end{aligned}, \tag{12}$$

where Gate_3 and Gate_4 are the weights corresponding to O_{md} and O_{sd}.

Then, we again aggregate the two types of fusion information Out_{h-w} and Out_{md-sd} to obtain the output of a single head Out_1,

$$\begin{aligned} \{\text{Gate}_5, \text{Gate}_6\} &= DPG\left(Cat\left(Out_{h-w}, Out_{md-sd}\right)\right) \\ Out_1 &= \text{Gate}_5 \odot Out_{h-w} + \text{Gate}_6 \odot Out_{md-sd} \end{aligned}, \tag{13}$$

where Gate_5 and Gate_6 are the weights corresponding to Out_{h-w} and Out_{md-sd}.

In the same way, we can obtain the features (Out_1, $\cdots$, Out_n) of all attention heads. As different heads reflect different features, that is to say, the contents in these maps have different weights. So, if we simply add or concatenate these heads directly, it can lead to information redundancy and even noise. To solve this problem. we introduce a learnable weight $\mathcal{W}$, which is iteratively computed as

$$\mathcal{W}^{(t+1)} = \mathcal{W}^{(t)} - \eta \cdot \frac{\partial \mathcal{L}}{\partial \mathcal{W}^{(t)}}, \tag{14}$$

where η is the learning rate, $\mathcal{L}$ represents the loss function for the current batch, and t is the number of training iterations. $\mathcal{W}$ is normalized via a softmax function to model their relative importance. The output of each head is then multiplied by its corresponding normalized weight, and the weighted head outputs are aggregated to produce the final output formulated as

$$\begin{aligned} \{\theta_1, \theta_2 \cdots \theta_n\} &= \text{softmax}\left(\mathcal{W}\right), \\ Out &= \sum_{i=1}^{n} \theta_i \odot Out_i, \end{aligned} \tag{15}$$

where θ_i denotes the i-th normalized dynamic fusion weight and n is the number of heads for multi-head attention.

4) Adaptive Feature Modulation (AFM): In order to improve the quality of feature fusion and the stability of training, we design an adaptive feature modulation (AFM), whose architecture is shown in Fig. 3. AFM first takes the fused advanced semantic feature Out as input and dynamically generates modulation parameters through DPG, namely $scale$ and $shift$. These two parameters are used to adjust the response strength and distribution offset of the feature channel, respectively. This process can be expressed as

$$\{Scale, Shift\} = DPG\left(Out\right), \tag{16}$$

where $Scale$ and $Shift$ denote the scaling factor and shift term, respectively. After obtaining the modulation parameters, AFM performs channel affine transformation on the original input feature F_{in}. Specifically, channel of F_{in} is multiplied by the corresponding scale factor, and then added to the shift term. Through this operation, important feature channels can be adaptively enhanced, while less informative responses can be recalibrated. The output feature of AFM (also AMPAM) is defined as

$$F_{out} = AFM\left(F_{in}, Out\right) = F_{in} \odot Scale + Shift. \tag{17}$$

It is worth noting that DPG serves as a general dynamic parameter generator, and its output dimension is determined by the specific module. For example, in DCP, each DPG generates a direction-specific spatial weight matrix to recalibrate the input feature before directional pooling. In MAPA, DPG is used as a gating function for adaptive feature fusion, and therefore generates two gate weights when two feature branches are fused. In AFM, DPG further generates the parameters of a conditional affine transformation, namely $Scale$ and $Shift$, which control the response intensity and distribution offset of the input feature, respectively. Therefore, although DPG adopts a similar generation structure in different modules, its outputs vary according to the specific application.

IV. Experimental Results and Analysis

A. Dataset Description

The effectiveness of AMPANet is evaluated on four datasets, namely UCM [55], AID [56], NWPU-RESISC45 [57] and MEET [1]. The dataset-specific training ratios follow widely used evaluation protocols for UCM, AID, and NWPU-RESISC45, reflecting their markedly different dataset sizes

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and difficulty levels. We randomly select 50 categories from the MEET dataset [1] and sample 1,000–2,000 images from each category to construct our MEET50 dataset. Accordingly, the reported results are intended to compare methods under the same dataset and split protocol, rather than to rank absolute performance across datasets with different class spaces and sample scales. These datasets have the following characteristics: they contain many natural and artificial scenes, the target scales are diverse, the resolution of the same ground object in different images is very different, and the spatial distribution density of targets in different scenes significantly changes. Their specific information is shown in Table I.

TABLE I
DETAILS OF FOUR AS SCENE DATASETS

Dataset	Total images	Scene classes	Image Per class	Image size	Spatial resolution
UCM	2100	21	100	256×256	0.3m
AID	10000	30	220-420	600×600	0.5-8m
NWPU-45	31500	45	700	256×256	0.2-30m
MEET50	63033	50	1000-2000	256×256 768×768 1280×1280	1m

B. Experimental Settings

In this study, we employed PyTorch as the underlying framework to implement all the selected state-of-the-art (SOTA) methods. For a fair comparison, all competing methods were reproduced under identical experimental conditions. For methods whose original papers specified the use of pre-training, ImageNet pre-trained weights were adopted strictly following the corresponding original settings. In addition, the same dataset splits, image preprocessing, and data augmentation strategies were consistently applied to all compared methods. The backbone of our approach is ResNet50 pre-trained on ImageNet [14]. All experiments were conducted on a workstation with 48 GB of RAM and a single 11 GB NVIDIA RTX 2080 Ti GPU. To evaluate the model, different training ratios (Tr) were set on four datasets: 20% and 50% of the images were randomly selected for training on the AID dataset, 10% and 20% on the NWPU-RESISC45 dataset, 80% on the UCM dataset and 10% on the MEET50 dataset, while the remaining images were used for testing. During training, we used the stochastic gradient descent (SGD) algorithm with a momentum of 0.9, weight decay of 0.001, and a label smoothing loss (smoothing factor is 0.1) [58], to optimize the network parameters. The initial learning rate was set to 0.01, with cosine annealing scheduling applied, and the batch size was set to 32. Image preprocessing and training-time augmentation included center cropping to 224×224, random horizontal and vertical flips, and random rotation. The overall accuracy (OA, which is defined as the ratio of correctly classified samples to the total number of samples, representing the average classification accuracy across all categories) and confusion matrix (CM, a specific table layout that visualizes the performance of a classification algorithm) [59] were used to evaluate the experimental performance. Each experiment was repeated 10 times with different random training/testing partitions.

C. Comparison Experiments

To comprehensively evaluate the effectiveness of our AMPANet, we compare it with thirteen representative state-of-the-art (SOTA) RSSC models on four widely used benchmark datasets. These selected SOTA RSSC methods are from different technical categories, such as MF2CNet [38], SAGN [40], GMFANet [39], EAM [43], DSDNet [42], and EMSCNet [44] focus on multi-scale or multi-level feature representation. CrossViT [18], Swin-T [45], and SGMSNet [49] are Transformer-based or CNN–Transformer hybrid methods for scene representation learning. MBLANet [50], EMA [36], HFAM [52], and MSCN [53] are attention-based or relational modeling methods. The comparison results are reported in Table II, where OA and standard deviation are used to quantitatively evaluate their superiority of accuracy. In addition, the number of parameters and GFLOPs are also listed to evaluate the trade-off between classification performance and computational cost. The specific analysis is presented as follows:

1) Experiments on the NWPU-RESISC45 Dataset: NWPU-RESISC45 is the most challenging of the three benchmarks due to its large number of scene categories and substantial intra-class variations. Table II shows that AMPANet achieves the highest OAs of 92.71% and 94.55% under the 10% and 20% training ratios, respectively. Among the multi-scale methods, MF2CNet obtains the best result under the 10% training ratio, reaching 92.07%, while EAM performs best under the 20% training ratio with an OA of 94.02%. AMPANet improves upon these results by 0.64% and 0.53%, demonstrating its ability to effectively integrate complementary information across different object scales while suppressing background interference. Compared with Transformer-based methods, AMPANet surpasses the strongest competitor, SGMSNet, by 0.25% and 0.07% under the 10% and 20% settings, respectively. This suggests that AMPANet preserves local discriminative details while modeling semantic dependencies, thereby alleviating the potential loss of spatial information caused by global feature modeling. Among the attention-based methods, MBLANet achieves competitive OAs of 92.59% and 94.33%, but AMPANet further improves them by 0.12% and 0.22%. Moreover, AMPANet outperforms HFAM by 0.20% and 0.25% while requiring only 5.17 GFLOPs, compared with 66.14 GFLOPs for HFAM. These results demonstrate that AMPANet achieves a favorable balance between classification accuracy and computational efficiency on this dataset.

2) Experiments on the AID Dataset: We can see the results of the AID dataset in Table II that, AMPANet achieves the highest OAs of 95.42% and 96.87% under the 20% and 50% training ratios, respectively. Among the multi-scale methods, EAM obtains the best results under both training ratios, reaching 95.02% and 96.46%, respectively. AMPANet surpasses EAM by 0.40% and 0.41%, and also outperforms MF2CNet by 0.88% and 0.85%. These improvements indicate that AMPANet not only captures multi-scale information but also emphasizes informative semantics and suppresses redundant features to produce more discriminative representations.

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TABLE II
COMPARISON WITH STATE-OF-THE-ART MODELS

Methods	Publication	UCM (%) ↑	AID (%) ↑		NWPU-RESISC45 (%) ↑		MEET50 (%) ↑	Params (M) ↓	GFLOPs (G) ↓	Peak Memory (MB) ↓	FPS (Images/s) ↑	GPU Hours (h) ↓	CE ↓
		80%	20%	50%	10%	20%	10%						
MBLANet [50]	TIP'2021	99.31 ± 0.19	95.14 ± 0.11	96.61 ± 0.09	92.59 ± 0.08	94.33 ± 0.09	82.58 ± 0.18	28.74	5.39	118.54	107.51	1.09	**19**
CrossVit [18]	ICCV'2021	98.99 ± 0.27	92.93 ± 0.28	95.29 ± 0.19	89.36 ± 0.07	91.61 ± 0.12	80.65 ± 0.27	26.50	21.10	**116.39**	59.11	1.58	29
Swin-T [45]	ICCV'2021	99.14 ± 0.23	94.55 ± 0.16	96.41 ± 0.23	91.95 ± 0.13	94.08 ± 0.13	82.15 ± 0.21	28.30	4.50	238.16	**180.19**	**0.66**	>50
MF2CNet [38]	TGRS'2022	99.01 ± 0.23	94.54 ± 0.15	96.02 ± 0.10	92.07 ± 0.13	93.89 ± 0.23	80.37 ± 0.15	33.20	**2.40**	157.28	31.01	3.05	32
EMA [36]	ICASSP'2023	99.07 ± 0.29	93.19 ± 0.20	96.26 ± 0.15	91.12 ± 0.18	93.70 ± 0.15	80.82 ± 0.30	23.80	5.03	118.35	50.02	1.76	30
SAGN [40]	TIP'2023	99.00 ± 0.29	94.44 ± 0.19	96.27 ± 0.24	91.44 ± 0.15	93.69 ± 0.13	81.26 ± 0.24	**23.33**	8.06	123.23	40.18	2.38	>50
EMSCNet [44]	TGRS'2023	99.06 ± 0.26	94.96 ± 0.17	96.23 ± 0.19	91.86 ± 0.19	93.99 ± 0.14	80.89 ± 0.19	85.83	16.85	383.76	115.43	1.14	42
HFAM [52]	TGRS'2024	99.19 ± 0.20	95.25 ± 0.13	96.58 ± 0.11	92.51 ± 0.13	94.30 ± 0.08	82.38 ± 0.13	25.57	66.14	237.37	70.87	1.08	27
GMFANet [39]	TGRS'2024	99.12 ± 0.28	94.19 ± 0.19	96.42 ± 0.20	89.92 ± 0.34	92.95 ± 0.15	81.73 ± 0.29	23.86	3.80	151.04	120.68	0.81	35
EAM [43]	NCA'2024	99.12 ± 0.24	95.02 ± 0.18	96.46 ± 0.17	92.01 ± 0.14	94.02 ± 0.13	82.34 ± 0.23	45.33	22.57	222.12	91.27	1.05	36
MSCN [53]	TGRS'2025	99.32 ± 0.27	94.51 ± 0.20	96.56 ± 0.13	90.36 ± 0.19	92.94 ± 0.14	81.65 ± 0.23	38.37	6.37	189.35	54.97	1.67	31
SGMSNet [49]	DAS'2025	99.17 ± 0.14	95.13 ± 0.18	96.35 ± 0.13	92.46 ± 0.15	94.48 ± 0.13	81.48 ± 0.18	34.67	5.50	158.86	21.48	4.05	43
DSDNet [42]	TGRS'2025	99.27 ± 0.11	94.26 ± 0.26	96.30 ± 0.25	91.88 ± 0.19	94.01 ± 0.17	82.02 ± 0.20	23.60	16.35	134.68	60.21	1.43	>50
Ours	-	**99.53 ± 0.11**	**95.42 ± 0.11**	**96.87 ± 0.12**	**92.71 ± 0.13**	**94.55 ± 0.14**	**82.67 ± 0.15**	33.29	5.17	156.02	54.86	1.65	28

For Transformer-based methods, AMPANet outperforms the strongest competitors, SGMSNet at the 20% training ratio and Swin-T at the 50% ratio, by 0.29% and 0.46%, respectively. More substantial improvements of 2.49% and 1.58% are observed over CrossViT. This demonstrates the advantage of combining semantic interactions with local structural information, particularly for scenes with similar global layouts. Among the attention-based methods, AMPANet surpasses HFAM by 0.17% under the 20% setting and MBLANet by 0.26% under the 50% setting. It also consistently outperforms MBLANet and EMA, confirming the effectiveness of its feature recalibration and relational interaction mechanisms.

3) Experiments on the UCM Dataset: As shown in Table II, most methods achieve OAs being close to or above 99% on UCM, indicating that classification performance on this dataset is nearly saturated. Nevertheless, AMPANet achieves the highest OA of 99.53%, demonstrating its strong feature discrimination capability. Among the multi-scale and multi-level methods, DSDNet performs best with an OA of 99.27%, followed by EAM and GMFANet with the same OA of 99.12%. AMPANet outperforms DSDNet, EAM, and GMFANet by 0.26%, 0.41%, and 0.41%, respectively. Such improvements remain meaningful near the performance upper bound, indicating that AMPANet can further refine discriminative representations even on a relatively less challenging dataset. Among the Transformer-based methods, SGMSNet achieves the highest OA of 99.17%, followed by Swin-T with 99.14%. AMPANet surpasses SGMSNet by 0.36%, showing that its combination of local details and semantic dependencies is more effective than general global modeling on UCM. For attention-based and relational modeling methods, MSCN, MBLANet, and HFAM achieve strong OAs of 99.32%, 99.31%, and 99.19%, respectively. AMPANet still outperforms MSCN by 0.21%, MBLANet by 0.22%, and HFAM by 0.34%. These results indicate that AMPANet can more effectively exploit fine-grained discriminative cues and reduce feature ambiguity between visually similar scenes through enhanced attention modulation and relational interaction.

4) Experiments on the MEET50 Dataset: MEET50 contains 50 categories sampled from MEET, with substantial variations in image size, and is evaluated using a 10% training ratio. As shown in Table II, AMPANet achieves the highest OA of 82.67% with a standard deviation of 0.15%. Among the multi-scale and multi-level methods, EAM performs best with 82.34%, while AMPANet surpasses it by 0.33 percentage points and improves upon MF2CNet by 2.30 percentage points. This suggests that adaptive multi-scale integration and semantic interactions enhance feature discrimination across image sizes. Among Transformer-based methods, Swin-T is the strongest competitor with 82.15%; AMPANet exceeds it by 0.52 percentage points and CrossViT by 2.02 percentage points, supporting the benefit of combining local details with contextual relations under limited training data. For attention-based and relational methods, AMPANet outperforms MBLANet and HFAM by 0.09 and 0.29 percentage points, respectively. Although the advantage over MBLANet is modest, its highest mean accuracy among all compared methods supports its effectiveness on MEET50.

5) Model Efficiency Analysis: To further evaluate model efficiency, Table II also reports Params, GFLOPs, peak memory, FPS, GPU hours, and CE for all compared methods. Peak memory and FPS are measured during inference on the same NVIDIA RTX 2080 Ti GPU with a batch size of 1 and FP32 precision. Peak memory is recorded using PyTorch CUDA memory statistics, while FPS is averaged over repeated forward passes after several warm-up iterations. GPU hours denotes the training time for one experiment, and Convergence Epoch (CE) is defined as the first epoch at which the validation performance is within 0.1 percentage points of the best achieved performance. Both GPU hours and CE are measured on the NWPU-RESISC45 dataset with the 20% training ratio under a 50-epoch training schedule. AMPANet requires 33.29 M parameters and 5.17 GFLOPs, with 156.02 MB peak memory, 54.86 FPS, 1.65 GPU hours, and a CE of 28. Although it does not have the smallest model size or the fastest inference speed, it achieves the best classification

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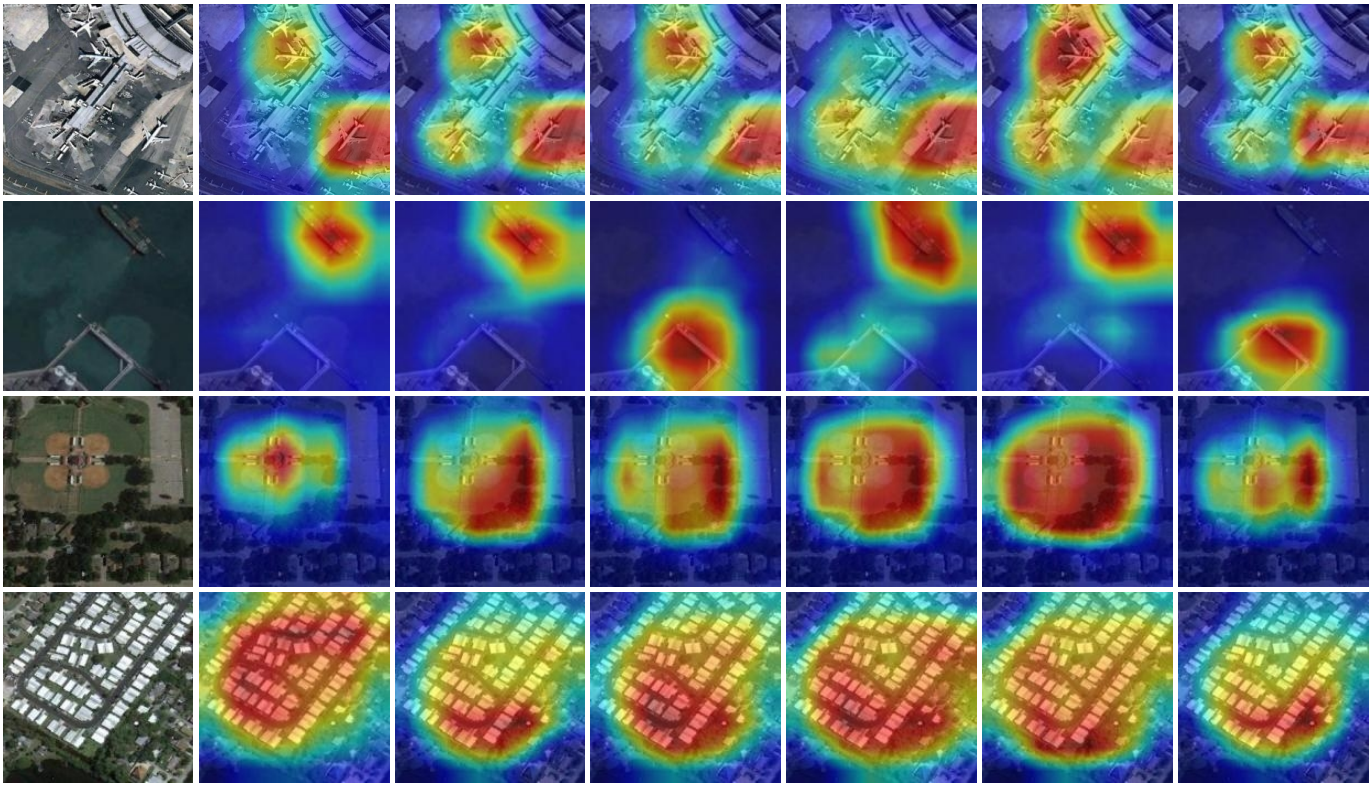

Fig. 6. Visual comparisons of heat maps produced by our network with different attentions, where images at the first column are the original images while those from the second column to the seventh column are generated by our AMPAM, CBAM [33], CA [17], ECA [34], SE [35], and EMA [36], respectively.

accuracy with moderate computational cost and convergence time, indicating a trade-off between accuracy and efficiency.

D. Ablation Studies

1) Comparison with Different Attention Mechanisms: We use Gradient-weighted Class Activation Map (Grad-CAM) [60] to visualize the last layer of the network, so as to observe the focus areas of the model and verify the superiority of our AMPAM. The visual results are shown in Fig. 6, which are generated by our network with these selected attention modules. From the results some observations can be concluded: (1) In the "Airplane" category of the first row, the target is mixed with objects of various shapes that have extremely similar gray levels. Our model is able to accurately find the two regions containing the target in the complex background, while all other models mistakenly regard the bottom left region as a region of interest. (2) In the "Ship" category of the second row, the target distribution in the scene is sparse, and "ResNet50 + CA", "ResNet50 + ECA", and "ResNet50 + EMA" mistake port facilities for ships, while our method obtains more accurate target areas than other methods. (3) In the "mobile-home-park" category of the fourth row, the objects are small, numerous and densely distributed, and our model can completely cover the actual object regions, while other methods only focus on the center of the scene and fail to pay enough attention to the upper object areas. The quantitative results are shown in Table III, which show that our model can more accurately focus on the key regions related to the scene semantics, whether it is a large-scale scene structure or a small-sized discriminative object, and is more beneficial for excluding interference and locating useful information in complex backgrounds. It can be inferred that our model has a more powerful feature extraction ability that takes both global and local features into account through the adaptive fusion mechanism.

TABLE III
COMPARISON OF THE EFFECTIVENESS OF METHODS

Method	UCM (%)	AID (%)		NWPU-RESISC45 (%)	
	80%	20%	50%	10%	20%
CBAM	99.09 ± 0.32	92.72 ± 0.26	96.23 ± 0.15	90.76 ± 0.20	93.69 ± 0.13
CA	99.00 ± 0.21	92.63 ± 0.24	96.18 ± 0.13	90.54 ± 0.29	93.67 ± 0.15
ECA	99.19 ± 0.24	93.46 ± 0.20	96.46 ± 0.12	91.03 ± 0.20	93.88 ± 0.10
SE	99.05 ± 0.24	92.65 ± 0.24	96.18 ± 0.14	91.06 ± 0.23	93.95 ± 0.09
EMA	99.07 ± 0.29	93.19 ± 0.20	96.26 ± 0.15	91.12 ± 0.18	93.70 ± 0.15
Ours	**99.53 ± 0.11**	**95.42 ± 0.11**	**96.87 ± 0.12**	**92.71 ± 0.13**	**94.55 ± 0.14**

Figure 7 presents their t-SNE visualization results, from which we can observe that, in the visualizations of SE and CA, there is a large mixed area of different colors highly interwoven, indicating that the features extracted by these classic attention mechanisms are not distinguished for some visually similar categories, causing the model to be unable to separate them in the feature space. In the visualizations of EMA and CBAM, many colored points are dispersed, which means that the features extracted by the model for images of the same category fluctuate greatly, indicating weak anti-interference ability. In contrast, our AMPAM exhibits clear gaps between categories, demonstrating that it can sensitively capture subtle differences between easily confused categories. Moreover, in the AMPAM visualization, almost all point

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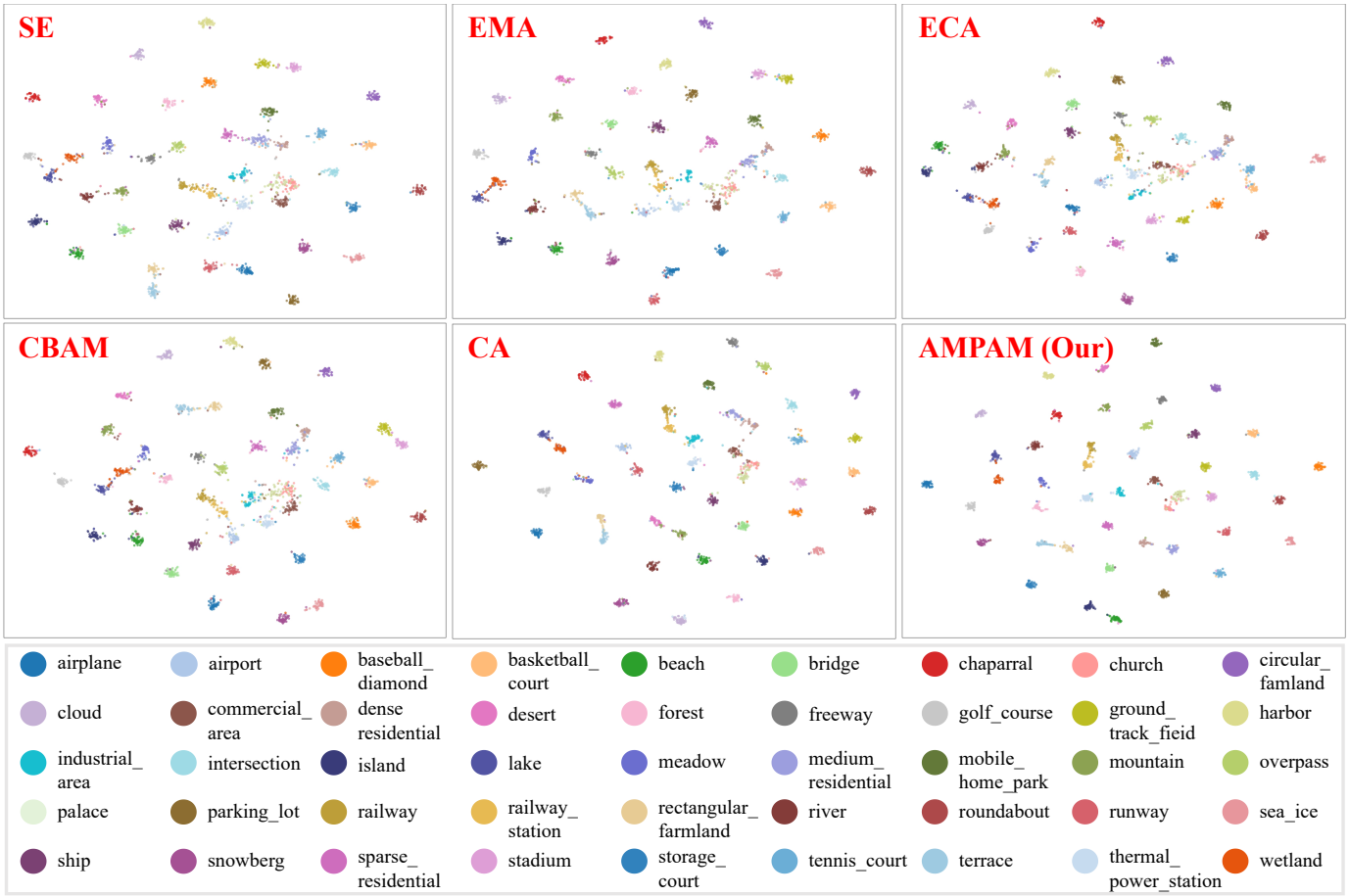

Fig. 7. T-SNE visual comparisons with different attentions including SE [35], EMA [36], ECA [34], CBAM [33], CA [17], and AMPAM (ours).

clusters shrink into high-density spheres, indicating that our AMPAM greatly suppresses background noise and viewpoint changes and ensures highly consistent feature representations for the same category.

TABLE IV
QUANTITATIVE OA COMPARISONS OF AMPAM WITH DIFFERENT BLOCKS.

Nets	UCM (%)	AID (%)		NWPU-RESISC45 (%)	
	80%	20%	50%	10%	20%
0	98.62 ± 0.35	92.67 ± 0.17	95.15 ± 0.20	89.15 ± 0.21	91.83 ± 0.19
1	98.81 ± 0.23	94.07 ± 0.15	95.74 ± 0.10	90.09 ± 0.13	93.89 ± 0.23
2	98.95 ± 0.16	94.48 ± 0.13	96.03 ± 0.08	90.99 ± 0.12	94.15 ± 0.13
3 (Ours)	**99.53 ± 0.11**	**95.42 ± 0.11**	**96.87 ± 0.12**	**92.71 ± 0.13**	**94.55 ± 0.14**
4	99.28 ± 0.12	95.13 ± 0.15	96.81 ± 0.10	92.70 ± 0.11	94.42 ± 0.10
5	99.43 ± 0.15	95.33 ± 0.13	96.33 ± 0.16	91.90 ± 0.20	94.22 ± 0.11
6	98.91 ± 0.14	94.17 ± 0.13	95.98 ± 0.09	90.32 ± 0.11	94.08 ± 0.15

2) Effectiveness of each block in AMPAM: To verify the effectiveness of each block in AMPAM, we categorize the experiments into the following types:

- **Net0:** Baseline (Resnet50).
- **Net1:** Baseline + MDWC + AFM.
- **Net2:** Baseline + DCP + MAPA + AFM.
- **Net3:** Baseline + MDWC + DCP + MAPA + AFM (ours).
- **Net4:** Baseline + MDWC (without DCKA) + DCP + MAPA + AFM.
- **Net5:** Baseline + MDWC (with regular convolution) + DCP + MAPA + AFM.
- **Net6:** Baseline + MDWC + DCP + MAPA + LN.

In the above types, Net0 uses ResNet50 as the baseline network for feature extraction. Net1 based on Net0 adds the "MDWC + AFM" structure after each residual block and downsampling block, thereby effectively capturing multi-scale features. Net2 based on Net0 with the "DCP + MAPA + AFM" structure effectively capture directional features. Net3 (our method) combines Net1 and Net2 using the "MDWC + DCP + MAPA + AFM" structure. And others are based on our architecture with different modules. Their testing subjective results are shown in Fig. 8 while objective ones are provided in Table IV. From the results, we can find that, with the help of the designed modules, both Net1 and Net2 perform better than Net0, indicating that the designed modules are effective. Furthermore, Net3 achieves the best performance with the usage of all blocks in both Net1 and Net2, demonstrating that the usage of all these modules can be more beneficial for extracting highlighted features (the fifth column in Fig. 8) and producing classification results. For Net4, we remove the adaptive structure (DCKA) in the MDWC module. From the visualization results shown in the sixth column in Fig. 8 and the OA in Table IV, Net4 is inferior to Net3, due to that it cannot adaptively adjust the weights of multi-scale extracted features, failing to focus on the important information. For Net5, we replace the depthwise separable

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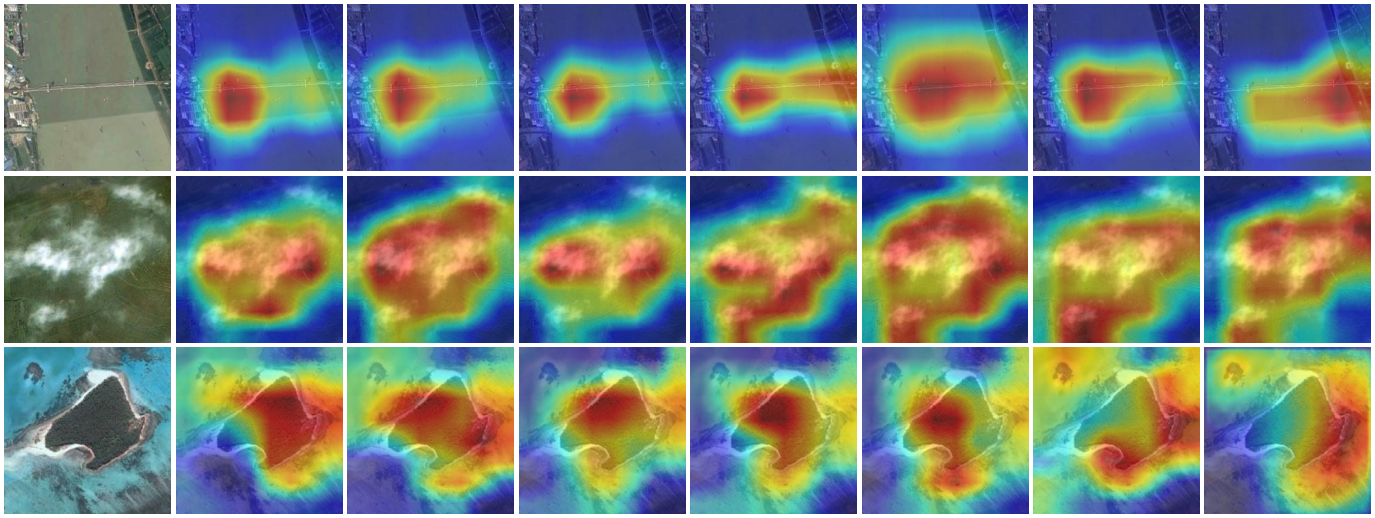
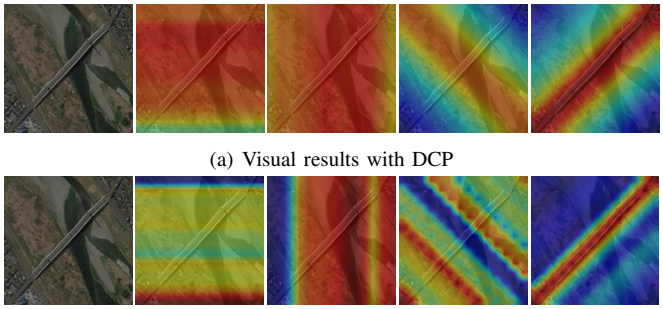


Fig. 8. Visual results with different nets shown in Table IV. Maps in the first column are the testing images while maps from the second column to the eighth column are produced by Net0, Net1, Net2, Net3, Net4, Net5, and Net6.

convolution with a standard convolution. The experimental results shown in the seventh column in Fig. 8 and the OA in Table IV illustrate that depthwise separable convolution is more helpful for our network to extract multi-scale features and improve classification performance. For Net6, AFM is replaced with LN. The experimental results shown in the eighth column in Fig. 8, especially a decrease in OA, indicate that AFM, compared to the static affine transformation of LN, can efficiently recalibrate the channel responses of the original features, thereby achieving higher model accuracy.

To further investigate the internal feature learning behavior of DCP and MAPA, we visualize the directional response maps generated by DCP and the corresponding attention distributions learned by MAPA. The visualization is performed on the AMPAM embedded in the last residual block of Layer3, where the feature maps retain sufficient spatial resolution for observing direction-dependent patterns. As shown in Fig. 9(a), DCP produces clearly different responses along the horizontal, vertical, main-diagonal, and secondary-diagonal directions. For the bridge scene, the horizontal and vertical branches mainly capture axis-aligned contextual structures, whereas the diagonal branches show more evident responses to oblique structures. In particular, the secondary-diagonal response is highly consistent with the main orientation of the bridge. These results intuitively demonstrate that DCP preserves complementary orientation-dependent spatial information rather than compressing the spatial structure into a single global representation.

As shown in Fig. 9(b), the four MAPA branches further exhibit distinct attention distributions. The horizontal and vertical attention maps present direction-dependent contextual responses, whereas the diagonal branches show stronger spatial correspondence with the oblique bridge structure. In particular, the secondary-diagonal attention is concentrated along the bridge region, indicating that MAPA can adaptively emphasize discriminative spatial regions by interacting the global contextual feature with direction-aware features. Since



(a) Visual results with DCP

(b) Visual results with MAPA

Fig. 9. Visual results with DCP and MAPA. The first column shows the test images, while the second to fifth columns show maps produced along the horizontal, vertical, main-diagonal, and secondary-diagonal directions.

the horizontal/vertical and diagonal features have different sequence lengths, the four MAPA attention maps are independently normalized for visualization. Therefore, they are used to illustrate the spatial attention distribution within each direction rather than to directly compare absolute attention magnitudes across different directions. Overall, these qualitative results reveal the complementary roles of DCP and MAPA: DCP explicitly extracts direction-aware spatial structures, while MAPA further selectively emphasizes discriminative information from these directional representations.

To further investigate the working mechanism of AFM, we analyze the learned modulation parameters and visualize the feature responses before and after modulation. Specifically, the actual $Scale$ and $Shift$ values are collected from the last AFM module of each stage over the entire test set, and their distributions are presented in Fig. 10. The learned $Scale$ values are mainly concentrated around 0.9134, while the $Shift$ values remain approximately 0.0865 across different stages. Moreover, neither parameter shows evident saturation toward the boundaries of the Sigmoid function. These relatively stable distributions suggest that AFM does not impose excessive perturbations on the original features. Instead, it performs

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moderate and input-dependent affine recalibration, allowing the network to adjust feature responses while largely preserving the underlying representation learned by the backbone.

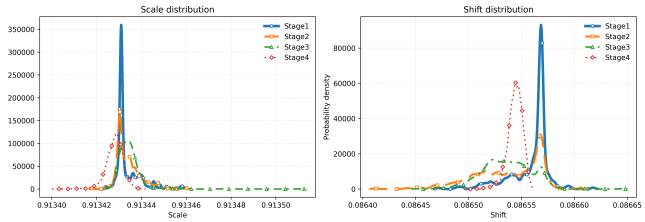

Fig. 10. Distributions of the learned $Scale$ and $Shift$ parameters at different stages over the entire test set.

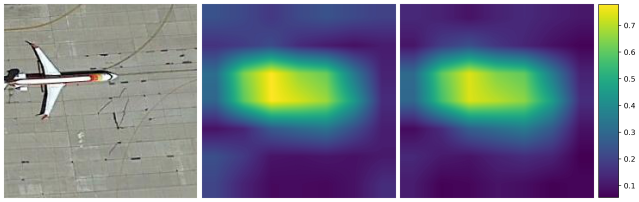

Fig. 11. Visualization of feature responses before and after AFM at Stage 4. From left to right: the original image, the feature response before AFM, and the feature response after AFM.

We further visualize the feature responses immediately before and after AFM at Stage4. For a fair comparison, the two feature maps are displayed using the same value range. As shown in Fig. 11, before modulation, relatively strong responses are distributed over both the airplane and several surrounding background regions, resulting in noticeable background interference. After AFM, the responses outside the target region are substantially weakened, while the dominant activation is retained and becomes more concentrated around the central airplane structure. Consequently, the distinction between the target-related response and the surrounding background becomes clearer. This observation indicates that the channel-wise adaptive scaling and shifting of AFM effectively recalibrates the relative importance of feature responses, thereby suppressing redundant background information while preserving task-relevant semantic cues. Together with the stable parameter distributions in Fig. 10, these results demonstrate that AFM achieves controlled feature refinement rather than aggressive modification of the backbone features.

3) Other Experiments: We further conduct additional ablation experiments on the NWPU-RESISC45 dataset using the 20% training ratio to validate the completeness of the proposed model. It is worth noting that when one factor is evaluated, the others are kept fixed. The analysis is presented as follows:

• **Analysis of Rotation Robustness** Although DCP employs four predefined pooling directions, remote sensing targets may appear at arbitrary orientations that are not aligned with these directions. To examine whether the predefined directions cause performance degradation for non-aligned targets, we conduct an additional rotation robustness experiment on NWPU-RESISC45. Specifically, the trained models are kept fixed, while the test images are rotated from 0° to 165° at 15° intervals. Three DCP configurations are evaluated, including horizontal–vertical pooling (DCP-HV), diagonal pooling (DCP-Diag), and the proposed four-direction pooling (DCP-4Dir). The mean OA and rotation standard deviation are calculated over all twelve rotation angles, while the maximum drop denotes the largest OA decrease relative to the 0° result.

TABLE V
ROTATION ROBUSTNESS COMPARISON AMONG DIFFERENT DCP CONFIGURATIONS.

DCP configuration	Mean OA (%) ↑	Rotation Std ↓	Worst-case OA (%) ↑	Maximum Drop (%) ↓
HV (0°, 90°)	93.99	0.195	93.82 (45°)	0.373
Diag. (45°, 135°)	94.09	0.176	93.97 (60°)	0.301
4Dir (Ours)	**94.41**	**0.147**	**94.29** (135°)	**0.267**

As reported in Table V, DCP-4Dir achieves the highest mean OA and the lowest rotation standard deviation. It also obtains the highest worst-case OA and the smallest maximum performance drop. Notably, the worst-case performance of DCP-4Dir occurs at 135°, which is itself one of the predefined pooling directions, rather than at a non-aligned orientation. This indicates that the performance variation is not systematically associated with the angular deviation from the predefined directions. Therefore, the four directions in DCP serve as complementary directional bases rather than restricting the representation to four discrete orientations. In addition, input-dependent spatial weighting within each directional branch and subsequent adaptive directional fusion enable complementary responses from multiple directions to represent targets at intermediate orientations. These results suggest that the fixed directional bases do not cause evident performance degradation for non-aligned targets, while providing stable directional representations without introducing additional learnable orientation parameters.

• **Verification in Complex Scenarios** To further evaluate the robustness of our model under practical challenging conditions, we conduct additional experiments involving class imbalance, label noise, and atmospheric artifacts. Specifically, different imbalance ratios and label noise levels are adopted, while cloud and haze perturbations with different severity levels are introduced to simulate atmospheric interference in remote sensing imagery. Class imbalance is simulated by class-wise subsampling according to the specified imbalance ratio, label noise is generated by randomly replacing a fixed proportion of training labels with incorrect labels, and cloud/haze degradation is applied to the test images at mild and moderate levels. The Baseline, SGMSNet, and our AMPANet are evaluated under the same experimental settings, and the quantitative results are reported in Table VI.

As shown in Table VI, the performance of all methods gradually decreases as the degree of class imbalance, label noise, or atmospheric degradation increases. Nevertheless, AMPANet consistently achieves the best results under all tested conditions, and its advantage becomes more evident under several more challenging settings, such as severe class imbalance and moderate haze. These results indicate that AMPANet exhibits stronger robustness to data-distribution shifts, annotation noise, and atmospheric interference. This advantage can be attributed to its joint modeling of multi-scale contextual information, directional spatial structures,

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TABLE VI
QUANTITATIVE OA COMPARISONS UNDER DIFFERENT PRACTICAL CHALLENGING CONDITIONS.

Methods	Class imbalance		Label noise			Cloud		Haze		Cross-dataset	
	IR=10	IR=20	10%	20%	30%	Mild	Moderate	Mild	Moderate	NWPU45→AID	AID→NWPU45
Baseline	84.04 ± 0.24	77.44 ± 0.27	88.78 ± 0.18	83.74 ± 0.23	77.19 ± 0.29	91.90 ± 0.17	86.86 ± 0.25	85.17 ± 0.28	62.67 ± 0.26	88.31 ± 0.21	77.11 ± 0.28
SGMSNet	87.72 ± 0.19	82.28 ± 0.23	90.74 ± 0.16	85.45 ± 0.21	79.65 ± 0.25	93.50 ± 0.14	90.31 ± 0.20	90.10 ± 0.22	74.40 ± 0.24	90.32 ± 0.17	80.49 ± 0.23
Ours	$\mathbf{88.16 \pm 0.15}$	$\mathbf{84.37 \pm 0.18}$	$\mathbf{90.82 \pm 0.13}$	$\mathbf{85.92 \pm 0.16}$	$\mathbf{79.83 \pm 0.20}$	$\mathbf{93.54 \pm 0.12}$	$\mathbf{90.47 \pm 0.15}$	$\mathbf{90.27 \pm 0.17}$	$\mathbf{77.21 \pm 0.21}$	$\mathbf{91.37 \pm 0.14}$	$\mathbf{81.69 \pm 0.19}$

and pairwise semantic relationships, which helps preserve discriminative features while suppressing irrelevant interference. Therefore, the proposed model shows better adaptability to challenging conditions beyond clean and class-balanced benchmark settings.

• **Cross-Dataset Generalization** To further evaluate the generalization ability of our model across different datasets, we conduct cross-dataset experiments between NWPU-RESISC45 and AID. Considering that the two datasets contain different scene categories, only the categories shared by both datasets are retained for evaluation. Specifically, the model trained on one dataset is directly evaluated on the other dataset without additional fine-tuning. The quantitative results are shown in Table VI. It can be observed that our method achieves the best performance in both the NWPU-45→AID and AID→NWPU-45 settings. These results demonstrate that the proposed AMPANet can maintain favorable classification performance under dataset distribution shifts, indicating its good cross-dataset generalization ability.

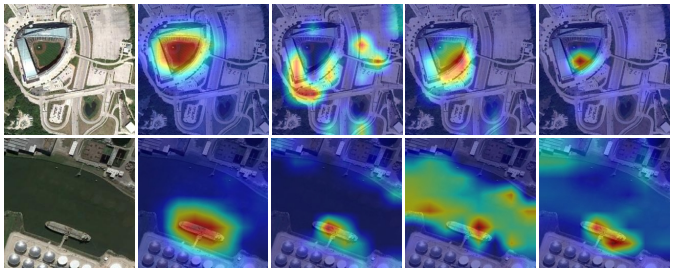

Fig. 12. Visual results compared with the selected YOLO methods. Maps in the first column are original images. Results from the second column to the fifth column are produced by our AMPAM, YOLOv8 [61], YOLOv11 [62], and YOLOv12 [63], respectively.

• **Comparison with object detection models.** To qualitatively compare the targeted feature localization capabilities of different models, we select YOLOv8 [61], YOLOv11 [62], and YOLOv12 [63] as representative classification models and use Grad-CAM [60] to visualize the activations of the last layer before their classification heads. The visual results for the two images are shown in Fig. 12. Our model produces more compact and target-aligned activation regions for both the "Stadium" and "Ship" categories. In contrast, the YOLO models exhibit varying degrees of scattered, incomplete, or diffuse activation, with some responses extending into surrounding roads, water, and other background regions. These results demonstrate that our model captures discriminative target features more effectively while suppressing irrelevant background responses. This advantage can be attributed to multi-scale and direction-aware feature extraction of our model, enhancing the representation of targets with diverse spatial structures and reducing interference from complex backgrounds. **More detailed hyperparameter tuning results are provided in the supplementary materials.**

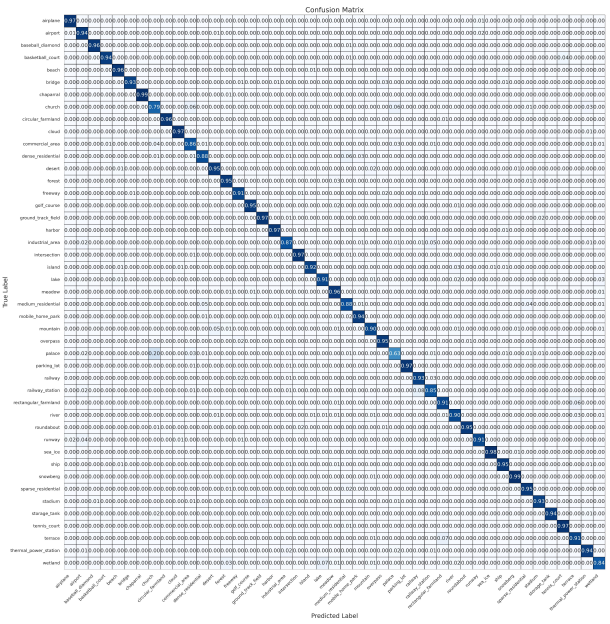

Fig. 13. Confusion matrix for the NWPU-RESISC45 dataset (Trs=10%).

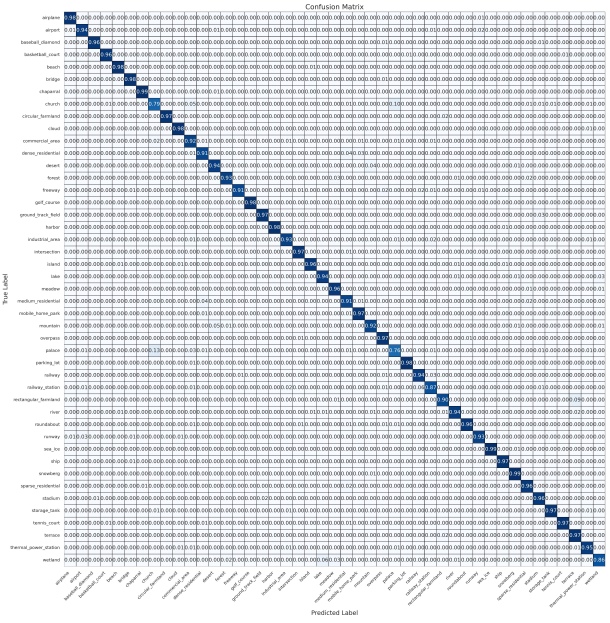

Fig. 14. Confusion matrix for the NWPU-RESISC45 dataset (Trs=20%).

E. Results of Confusion Matrix

Figures 13 and 14 present the normalized confusion matrices obtained on the NWPU-45 dataset using training ratios

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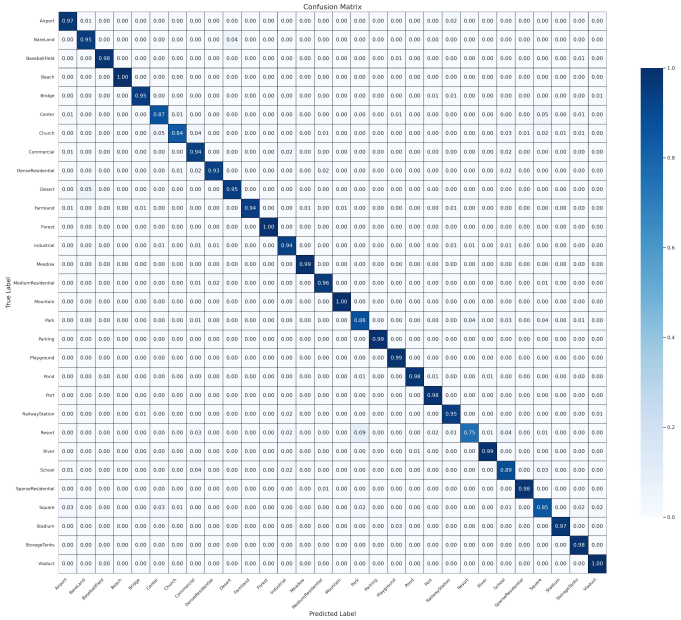

Fig. 15. Confusion matrix for the AID (Trs=20%).

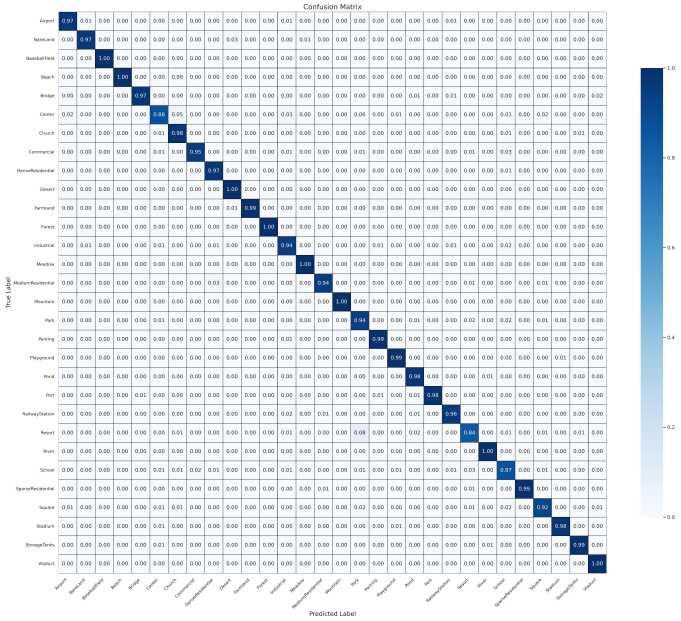

Fig. 16. Confusion matrix for the AID (Trs=50%).

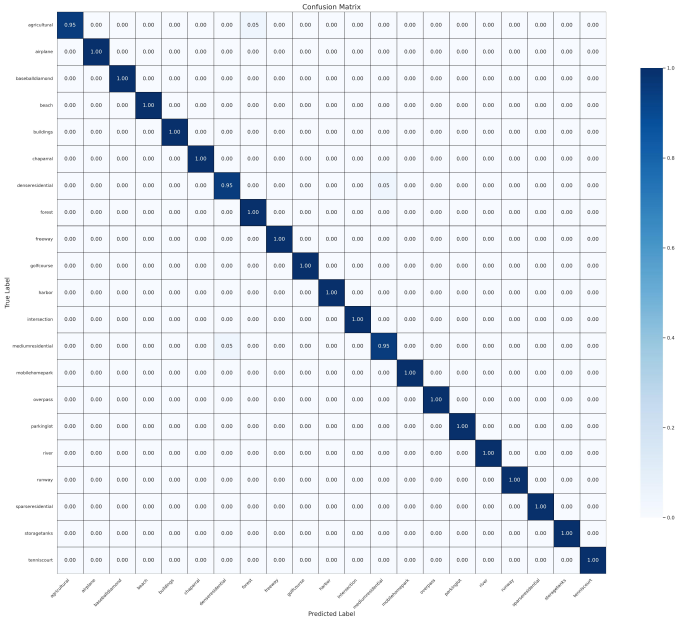

Fig. 17. Confusion matrix for the UCM dataset (Trs=80%).

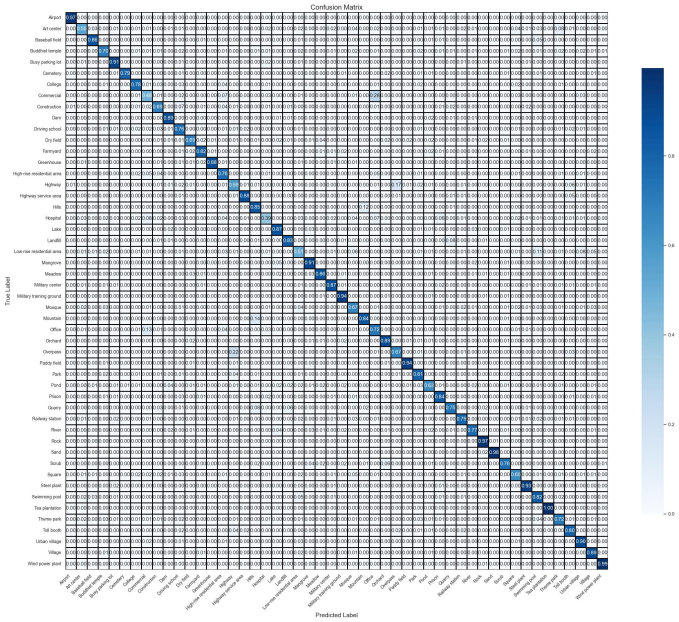

Fig. 18. Confusion matrix for the MEET50 dataset (Trs=10%).

of 10% and 20%, respectively. The results exhibit a strong diagonal distribution, demonstrating the excellent class-wise recognition capability of the proposed model. Specifically, 37 and 41 out of the 45 scene categories achieve accuracies of at least 90% under the two training settings, respectively. As the training ratio increases, the classification performance of most categories is further improved.

Figures 15 and 16 show the normalized confusion matrices obtained on the AID dataset using training ratios of 20% and 50%, respectively. At the 20% training ratio, 23 of the 30 categories achieve accuracies of at least 90%; this number further increases to 27 when the training ratio reaches 50%. Notably, several challenging categories show substantial improvements with additional training samples. For instance, the accuracies of "Church", "Park", and "Square" increase from 0.84, 0.88, and 0.85 to 0.98, 0.94, and 0.92, respectively. Moreover, several categories, including "Beach", "Forest", "Mountain", and "Viaduct", are perfectly classified. Most scene categories are accurately distinguished, with only occasional misclassification between visually similar categories such as "Center" and "Square". The non-trivial off-diagonal confusion between "Center" and "Square" in AID can be mainly attributed to their shared urban morphology: both categories often contain open paved areas, regular road intersections, dense surrounding buildings, and highly geometric spatial layouts. This case indicates that although AMPANet effectively captures multi-scale contextual and pairwise relational cues, distinguishing fine-grained urban functional semantics remains challenging when global layouts and local structural elements are highly

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TABLE VII
MCNEMAR TEST RESULTS UNDER DIFFERENT DATASETS AND TRAINING RATIOS.

Methods	UCM (%)			AID (%)						NWPU-RESISC45 (%)					
	80%			20%			50%			10%			20%		
	b	c	p_{Mc}	b	c	p_{Mc}	b	c	p_{Mc}	b	c	p_{Mc}	b	c	p_{Mc}
MBLANet	7	1	0.0703	181	121	< 0.001	61	37	0.0197	736	609	< 0.001	528	413	< 0.001
MF2CNet	8	1	0.0391	226	114	< 0.001	82	47	0.0026	896	707	< 0.001	645	496	< 0.001
SGMSNet	8	1	0.0391	162	124	0.0285	73	48	0.0291	762	663	0.0094	638	466	< 0.001
HFAM	7	0	0.0156	353	135	< 0.001	69	46	0.0402	799	667	< 0.001	573	500	0.0279
EAM	6	0	0.0312	185	131	0.0029	77	50	0.0210	863	717	< 0.001	543	469	0.0217

similar.

Figure 17 presents the normalized confusion matrix obtained on the UCM dataset using an 80% training ratio. The proposed model achieves 100% classification accuracy in 18 of the 21 scene categories. The remaining three categories, namely "Agricultural", "Dense Residential", and "Medium Residential", all achieve an accuracy of over 0.95. Only a few samples are misclassified between visually related categories. The diagonal confusion matrix demonstrates the outstanding classification performance of the proposed model and indicates that it achieves near-saturated performance on the UCM dataset under this experimental setting.

Figure 18 shows the normalized confusion matrix on MEET50 using a 10% training ratio. The matrix is predominantly diagonal, with 11 of the 50 categories attaining displayed accuracies of at least 90%. "Tea plantation" reaches 1.00, while "Wind power plant", "Sand", and both "Airport" and "Rock" achieve 0.99, 0.98, and 0.97, respectively, indicating effective recognition of scenes with characteristic textures or spatial structures. However, "Hospital", "Commercial", and "Low-rise residential area" obtain only 0.39, 0.48, and 0.51, respectively. Confusion from "Commercial" to "Office" reaches 28%, compared with 13% in the reverse direction; the corresponding rates between "Highway" and "Overpass" are 17% and 22%, respectively. These errors may reflect shared urban building patterns and similar road structures, for which local appearance provides limited discriminative cues. The results support effective recognition of many categories under limited training data, while distinguishing visually similar scenes with different functional semantics remains challenging.

F. Significance Test Analysis

The OA comparisons in Table II show that AMPANet achieves competitive performance on the three benchmark datasets. To further verify whether these improvements are statistically reliable, we compare AMPANet with five strong competitors whose OAs are close to ours, including MBLANet, MF2CNet, SGMSNet, HFAM, and EAM. Two types of significance analysis are conducted in this section. The McNemar test is used to evaluate sample-level prediction differences between two classifiers, while the paired t-test and Wilcoxon signed-rank test are used to evaluate the stability of OA differences over 10 repeated runs. In all tests, $p < 0.05$ indicates statistical significance.

1) McNemar Test: The McNemar test focuses on the discordant predictions of two classifiers on the same test set. Let b be the number of samples correctly classified by AMPANet but incorrectly classified by the compared method, and let c be the number of samples incorrectly classified by AMPANet but correctly classified by the compared method. The null hypothesis is that the two classifiers have the same error distribution, namely $b = c$. Let $n = b + c$ denote the number of discordant pairs. Under the null hypothesis, the number of one-sided discordant samples follows a binomial distribution, *i.e.*, $X \sim \mathrm{Binomial}(n, 0.5)$. Therefore, the exact binomial McNemar p-value can be written as

$$\begin{aligned} p_{\mathrm{Mc}} &= \min\{1, 2\Pr(X \le \min(b,c))\} \\ &= \min\left\{1, 2\sum_{i=0}^{\min(b,c)} \binom{n}{i}\left(\frac{1}{2}\right)^n\right\}, \end{aligned} \tag{18}$$

where $\Pr(\cdot)$ denotes the probability of the event inside the parentheses, $\binom{n}{i}$ denotes the binomial coefficient, *i.e.*, choosing i samples from the n discordant samples.

From Table VII, most comparisons satisfy $p < 0.05$, demonstrating that AMPANet has statistically significant sample-level advantages in most settings. On UCM, the comparison with MBLANet is not significant ($p = 0.0703$), mainly because UCM is close to performance saturation and only a few discordant samples are produced. On AID and NWPU-RESISC45, AMPANet achieves significant differences against all five competitors under both training ratios, indicating that its improvements are not caused by occasional prediction fluctuations.

2) Paired t-test and Wilcoxon Signed-rank Test: To evaluate the stability of AMPANet over repeated experiments, we further perform the paired t-test and Wilcoxon signed-rank test on the OA results from 10 repeated runs. For each dataset and training ratio, AMPANet and each compared method are paired according to the same run index. Let x_i and y_i denote the OA values of AMPANet and a compared method in the i-th paired run, respectively. The paired difference is defined as

$$d_i = x_i - y_i, \quad i = 1, 2, \ldots, n, \tag{19}$$

where n is the number of valid paired runs. The paired t-test evaluates whether the mean of d_i is significantly different from zero, while the Wilcoxon signed-rank test evaluates whether the paired differences are symmetrically distributed around zero without assuming normality. Since multiple methods are compared under the same dataset and training ratio, the raw probabilities (p-values) are corrected using the Holm-Bonferroni procedure. In Table VIII, T_p and W_p denote the Holm-adjusted p-values of the paired t-test and Wilcoxon

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TABLE VIII
WILCOXON SIGNED-RANK TEST AND PAIRED t-TEST RESULTS OVER 10 REPEATED RUNS.

Methods	UCM			AID						NWPU-RESISC45					
	80%			20%			50%			10%			20%		
	T_p	W_p	Sign.	T_p	W_p	Sign.	T_p	W_p	Sign.	T_p	W_p	Sign.	T_p	W_p	Sign.
MBLANet	0.0310	0.0273	Yes	< 0.001	0.0098	Yes	0.0010	0.0098	Yes	0.0359	0.0488	Yes	0.0121	0.0195	Yes
MF2CNet	0.0014	0.0098	Yes	< 0.001	0.0098	Yes	< 0.001	0.0098	Yes	< 0.001	0.0098	Yes	< 0.001	0.0098	Yes
SGMSNet	< 0.001	0.0098	Yes	0.0016	0.0109	Yes	< 0.001	0.0098	Yes	0.0038	0.0117	Yes	0.2873	0.2754	No
HFAM	0.0012	0.0098	Yes	< 0.001	0.0098	Yes	< 0.001	0.0098	Yes	0.0201	0.0098	Yes	0.0013	0.0098	Yes
EAM	0.0310	0.0273	Yes	< 0.001	0.0098	Yes	0.0010	0.0098	Yes	< 0.001	0.0098	Yes	< 0.001	0.0098	Yes

signed-rank test, respectively, and Sign. indicates whether $W_p < 0.05$, i.e., whether the difference is significant.

As shown in Table VIII, most Holm-adjusted p-values are smaller than 0.05, confirming that AMPANet is statistically stable over repeated trials. The only non-significant case occurs when AMPANet is compared with SGMSNet on NWPU-RESISC45 using 20% training samples, where the OA gap is only 0.07%. This indicates that SGMSNet is a very competitive baseline in this setting, while the overall results still support the robustness and effectiveness of AMPANet.

G. Failure Cases

Although our method had achieved marketable progress in performance improvement, some confusion remains between categories with highly similar visual characteristics, such as "Palace" and "Church". As further shown by the failure-case visualizations in Fig. 19, the misclassified "Church" and "Palace" samples both contain large central buildings, regular roof structures, surrounding courtyards, roads, and vegetation. The activation regions are mainly concentrated on these shared architectural patterns, while category-specific details are relatively weak from the remote-sensing perspective. Therefore, the root cause of these errors is not that the model completely fails to locate the main semantic objects, but that the discriminative cues between the two categories are overwhelmed by highly similar object appearances and spatial layouts. Therefore, further improving the model's ability to distinguish highly similar scene samples will be one of the focuses of our future work.

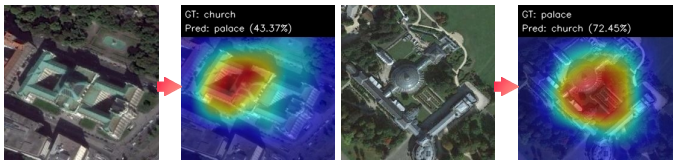

Fig. 19. Visualization of representative failure cases for "Church" and "Palace".

V. CONCLUSIONS

To cope with the challenges of substantial object-scale variations, complex spatial layouts, and strong visual similarities among different scene categories in remote sensing scene classification, this paper proposes an Adaptive Multi-scale Pairwise Attention Network (AMPANet) to enhance scene representations through multi-scale feature learning, direction-aware context modeling, and pairwise semantic interaction. To be specific, in AMPANet, four key components are associated to address these issues to achieve robust and accurate classification performance, such as Multi-Scale Depthwise Separable Convolution (MDWC) is to efficiently capture features at multiple spatial scales, Dynamic Coordinate Pooling (DCP) is to encode complementary directional information, Multi-Head Adaptive Pairwise Attention (MAPA) explicitly models interactions between semantic regions, Adaptive Feature Modulation (AFM) adaptively integrates the resulting relational features and suppresses redundant information. In experiments, we first verified the effectiveness of the selection of each module and hyper-parameters, especially our MAPA. Then, our AMPANet was experimented on the UCM, AID, NWPU-RESISC45 and MEET50 datasets, and achieved overall accuracies of 82.67% on meet50, 99.53% on UCM, 95.42% and 96.87% on AID, and 92.71% and 94.55% on NWPU-RESISC45, which is more competitive to the representative selected SOTA methods. The results of confusion matrix validated that our AMPANet also performed well on the classification of categories with highly similar appearances and semantic layouts. The significance tests further confirm that the improvements are statistically reliable in most experimental settings. Beyond scene classification, the proposed multi-scale pairwise attention design also shows promising potential for a broader range of remote sensing interpretation tasks. For object detection, AMPANet can be integrated into backbone networks or feature pyramid architectures to strengthen object-level relational modeling under substantial scale variations and complex backgrounds. For semantic segmentation, its direction-aware context modeling and adaptive feature modulation can help preserve boundary-sensitive spatial details while effectively incorporating long-range contextual information. Accordingly, extending AMPANet to object detection, semantic segmentation, and other dense prediction tasks is a promising direction for further enhancing its applicability and practical value in remote sensing image understanding.

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Zhenghua Huang (Member, IEEE) received his Ph.D. degree from the School of Artificial Intelligence and Automation, Huazhong University of Science and Technology (HUST), Wuhan, in 2017. He is currently an Associate Professor with the School of Computer Science and Technology, Wuhan University of Science and Technology, Wuhan. He was selected as one of the top 2% global scientists in 2020, 2021, and 2023. In the past two years, he has reviewed over 600 SCI papers in IEEE TIP/TCSVT/TGRS/TCYB/TMM, and so on, and was selected as one of the best reviewers of IEEE TGRS in 2025 and a Topical Associate Editor of IEEE TGRS in 2026. His research interests include computer vision, deep learning, image processing, and optical flow estimation.

Yingjie Zhang received his B.S. degree from the School of Electrical and Information Engineering, Wuhan Institute of Technology, Wuhan, China, in 2023. He is currently pursuing the master's degree at the School of Electrical and Information Engineering, Wuhan Institute of Technology, Wuhan. His research interests include image classification, deep learning, and computer vision.

Qian Li received her M.S. degree from the College of Biomedical Engineering, South Central University For Nationalities, Wuhan, China, in 2016. She is a lecturer with the School of Electrical Information Engineering, Wuhan Donghu University, Hubei, China. Her research interests include image processing, deep learning, and computer vision.

Yaozong Zhang received his Ph.D. degree from the School of Artificial Intelligence and Automation, Huazhong University of Science and Technology, Wuhan, China, in 2018. He is currently a Lecturer with the Electrical and Information Engineering School, Wuhan Institute of Technology, Hubei, China. His research interests include remote sensing image (hyperspectral image and SAR image) degradation correction, multimodal image registration, and target detection and recognition.

Chunhua Deng received his Ph.D. degree in pattern recognition and intelligent systems from the School of Artificial Intelligence and Automation, Huazhong University of Science and Technology (HUST), Wuhan, China, in 2016. He is currently an Associate Professor with the School of Computer Science and Technology, Wuhan University of Science and Technology, Wuhan. His research interests include computer vision, pattern recognition, and machine learning.

Xin Xu (Senior Member, IEEE) received his B.S. and Ph.D. degrees in computer science and engineering from Shanghai Jiao Tong University, China, in 2004 and 2012, respectively. He is currently a Full Professor with the School of Computer Science and Technology, Wuhan University of Science and Technology, China. He was shortlisted as the Best Paper Finalist of the IEEE International Conference on Multimedia and Expo (ICME) in 2021. His research interests include artificial intelligence, computer vision, and image processing.

Yansheng Li (Senior Member, IEEE) received the B.S. degree in information and computing science from Shandong University, Weihai, China, in 2010, and the Ph.D. degree in pattern recognition and intelligent system from the Huazhong University of Science and Technology, Wuhan, China, in 2015.

He is currently a Professor with the School of Remote Sensing and Information Engineering, Wuhan University (WHU), Wuhan. From 2017 to 2018, he was a Visiting Assistant Professor with the Department of Computer Science, Johns Hopkins University, Baltimore, MD, USA. He has authored more than 100 peer-reviewed journal and conference papers. His research interests mainly lay in the field of knowledge graph, deep learning, and their applications in remote sensing big data mining.

He is an Associate Editor of IEEE TGRS, a Junior Editorial Member of The Photogrammetric Record, and a Lead Guest Editor of IEEE JSTARS and Remote Sensing.

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